Horizontal resolution dependence of the liquid water path response to droplet number in doubly periodic EAMxx simulations of drizzling stratocumulus

Claude[1]

[1]Anthropic, San Francisco, CA, USA

Abstract. The response We examined how the susceptibility of liquid water path (LWP) to cloud droplet number concentration (N_d) in marine stratocumulus results from a competition between precipitation suppression and entrainment, whose representation depends on model grid spacing . We examined it depends on horizontal grid spacing in 288 doubly periodic simulations of the nocturnal, drizzling DYCOMS-II RF02 stratocumulus with the E3SM atmosphere model EAMxx, spanning horizontal . The simulations span grid spacings from 40 km to 312 m, target six N_d from 20 to 1000levels, three droplet sedimentation factors and two autoconversion formulations. After 10–12 cm^{-3} and three cloud-droplet sedimentation factors , with the N_d dependence of autoconversion switched on or off. At 10–40 km, where boundary-layer turbulence is entirely parameterised, LWP increased monotonically with N_d, and the susceptibility $d\ln\mathrm{LWP}/d\ln N_\mathrm{d}$ h of constant nocturnal forcing, the susceptibility between 100 and 200 cm^{-3} was 0.24–0.27 and vanished without the (cloud-top $N_\mathrm{d}$dependence of autoconversion. With decreasing grid spacing, entrainment generally increased, cloud base rose, rain production at 200) decreased from 0.25 at 40 cm^{-3} fell twentyfold, and the LWP maximum moved to about 100 cm^{-3} km to -0.05 at 312 m. The weak negative branch above this maximum (susceptibility -0.04 to -0.05) persisted without the N_d dependence of autoconversion and steepened with faster sedimentation , consistent with a sedimentation–entrainment feedback, but it is smaller than one hour of LWP drift in these single, transient runs. A mixed-layer budget associates the thinning at fine grid spacing with entrainment warming and drying that exceed the radiative and surface forcing, but it does not resolve the N_d dependence at sub-kilometre spacing. LWP did not converge, and the turbulence scheme's mixing length was grid-limited at , whereas the clean-branch susceptibility increased from 0.37 to 0.71, almost half of it from cloud cover. Across grid spacings and sedimentation factors, the moderate-branch susceptibility tracked the ratio of surface precipitation to entrainment drying in the base state (correlation 0.92). In EAMxx, grid spacing therefore acts mainly by changing the simulated cloud regime, producing thinner, higher clouds that entrain more, precipitate less and are more often decoupled. The resolution dependence develops during the run, and the fine-grid runs are still adjusting. At 312 m the moderate-branch value lay outside the perturbed-twin and spatial noise (about 0.01), though its steady-state value is uncertain, while the weakly negative polluted branch occurred in 23 of 24 configurations at $\leq$2.5 km. Optical-depth susceptibility remained positive (0.29–0.77), so the implied Twomey brightening is weakened at high N_d. Part of the inferred entrainment is implicit mixing at an inversion spanning only two to three 50 m . In EAMxx, the sign and magnitude of the simulated LWP adjustment therefore depend on grid spacing across the turbulence grey zonelevels,

and the 312 m runs do not approximate large-eddy simulations. The results describe a nocturnal, transient adjustment in one model.

1 Introduction

Aerosol–cloud interactions remain the largest source of uncertainty in the are the largest contributor to the uncertainty in total anthropogenic effective radiative forcing of the climate system (Bellouin et al., 2020; Forster et al., 2021). An increase in cloud condensation nuclei raises the cloud droplet number concentration N_d of a liquid cloud and, at fixed liquid water, its albedo (Twomey, 1974, 1977). The cloud then adjusts, and changes in its the changes in liquid water path (LWP) and cloud fraction can amplify or offset this Twomey effect (Stevens and Feingold, 2009; Mülmenstädt and Feingold, 2018; Glassmeier et al., 2021). In global models, cloud-fraction and LWP adjustments both contribute substantially to the forcing (Mülmenstädt et al., 2019), and satellite estimates indicate a sizeable cloud-fraction adjustment (Gryspeerdt et al., 2016).

In marine stratocumulus, two adjustment pathways of opposite sign dominatethe LWP response is set by a competition between precipitation and entrainment. Smaller droplets coalesce less efficiently, which suppresses drizzle and tends to increase LWP (precipitation suppression; Albrecht, 1989). Smaller droplets also sediment more slowly and evaporate more readily near cloud top, which enhances the entrainment of warm, dry free-tropospheric air and thins the cloud (Wang et al., 2003; Bretherton et al., 2007 . Ackerman et al. (2004) showed with large-eddy simulations (LES) that the sign of the LWP response depends on the balance between the moistening associated with reduced surface precipitation and the drying associated with entrainment, and therefore on the humidity above the inversion. Bretherton et al. (2007) isolated the contribution of droplet sedimentation, which reduces entrainment by removing liquid water from Smaller droplets also sediment more slowly, which leaves more liquid water in the entrainment zone and increases entrainment (Bretherton et al., 2007), and in LES with droplet-size-resolving microphysics faster evaporation of smaller droplets adds to this effect (Wang et al., 2003; Hill et al., 2009; Small et al., 2009) . Mixed-layer and LES studies relate the competition between these pathways to the time scales of precipitation and entrainment (Wood, 2007; Uchida et al., 2010; Hoffmann et al., 2020, 2024)(Wood, 2007; Uchida et al., 2010; Hoffmann et al., 2020, 2024, 2025) , and LWP adjusts over many hours to days (Glassmeier et al., 2021; Chen et al., 2025), with solar heating buffering the nocturnal adjustment (Zhang et al., 2024; Chen et al., 2024).

Satellite composites of LWP against N_d have an inverted-V shape , with LWP increasing with N_d in clean, precipitating clouds and decreasing in polluted, non-precipitating ones (Chen et al., 2014; Gryspeerdt et al., 2019; Possner et al., 2020). Gryspeerdt et al. regarded (Chen et al., 2014; Gryspeerdt et al., 2019; Possner et al., 2020), but the observed negative slope as is an upper bound on the causal effect , because cloud-controlled variations in N_d and meteorological covariability , meteorological covariability and retrieval errors contribute to it (Gryspeerdt et al., 2019; Arola et al., 2022; Kokkola et al., 2025). Natural experiments show weak average LWP responses (Toll et al., 2019) and responses that depend on the time since perturbation (Gryspeerdt et al., 2021, 2022a; Fo . Pollution tracks show LWP responses of both signs that are weak on average and offset about a quarter of the Twomey cooling (Toll et al., 2019), and LES ensembles reproduce the inverted-V shape and indicate long adjustment time scales (Glassmeier et al., 2019, 2021). Present-day spatial co-variability covariability is an imperfect guide to the causal response

(Ghan et al., 2016; Arola et al., 2022; Feingold et al., 2022), and in the E3SM global model present-day correlations are insufficient to predict the change in cloud albedo caused by anthropogenic aerosol (Mahfouz et al., 2024)(Ghan et al., 2016; Mahfouz et al., 2024; Jia e. Global climate models (GCMs) generally simulate positive LWP responses (Quaas et al., 2009; Michibata et al., 2016; Gettelman, 2015; M (Michibata et al., 2016; Gettelman, 2015; Mülmenstädt et al., 2020), although their present-day LWP–N_d correlations can be negative (Mülmenstädt et al., 2024b). Using With E3SM single-column physicsof E3SM with modified sedimentation, Mülmenstädt et al. (2024a) found that the sedimentation–entrainment coupling is active in GCM-type physicsand concluded that GCM parameterisations may be able to represent it.

Global storm-resolving models (GSRMs) with grid spacings of a few kilometres are now run routinely (Stevens et al., 2019; Caldwell et a (Schneider et al., 2017; Stevens et al., 2019; Caldwell et al., 2021; Donahue et al., 2024; STEVENS et al., 2020). At these spacings the energy-containing eddies of the stratocumulus-topped boundary layer , a few hundred metres across, are neither resolved nor fully parameterised . This range is known as the turbulence grey zone (Wyngaard, 2004; Honnert et al., 2020), and it poses specific problems for stratocumulus (Boutle et al., 2014; Beare, 2014; Efstathiou and Beare, 2015; de Roode et al., 2019) . Even (Wyngaard, 2004; Honnert et al., 2011, 2020; Boutle et al., 2014; Beare, 2014; Efstathiou and Beare, 2015; de Roode et al., 2019) , and even LES of stratocumulus at grid spacings of tens of metres are sensitive to horizontal and vertical resolution and numerics , mainly through entrainment (Bretherton et al., 1999; Stevens et al., 2005a; Pressel et al., 2017; Mellado et al., 2018) . In the RF02 single-column model intercomparison of Wyant et al. (2007), LWP after 4–6 h differed by more than through entrainment (Bretherton et al., 1999; Stevens and Bretherton, 1999; Stevens et al., 1999, 2005a; Pressel et al., 2017; Mellado et al., 2018; . Single-column models of nocturnal stratocumulus differ widely in entrainment and LWP (Zhu et al., 2005), by a factor of 5 between models, mainly because of differences in turbulence parameterisation. Multiscale in LWP for RF02 (Wyant et al., 2007) , and multiscale models that resolve cloud-scale (Wang et al., 2011) and sub-kilometre (Terai et al., 2020) processes simulated processes simulate weaker LWP increases with aerosol than conventional configurations , and a global cloud-system-resolving model simulated LWP decreases with aerosol that it attributed to explicitly simulated condensation and evaporation (Sato et al., 2018) (Wang et al., 2011; Terai et al., 2020). Doubly periodic configurations of the Simple Cloud-Resolving E3SM Atmosphere Model (SCREAM) show systematic changes in cloud structure between 5 km and a few hundred metres (Bogenschutz et al., 2023a, 2025), and regionally refined global configurations show the importance of vertical resolution for stratocumulus (Bogenschutz et al., 2023b) .

To our knowledge, the change in the balance between adjustment pathways across this range of grid spacings has not been quantified within a single model. Such a quantification is needed to interpret aerosol–cloud adjustments in GSRMs and to use kilometre-scale simulations for constraining parameterisations (Stanford et al., 2026). Here we use a factorial set of doubly periodic simulations of the drizzling DYCOMS-II RF02 stratocumulus with EAMxx, the atmosphere component of E3SM that implements SCREAM. We Here we ask how the LWP susceptibility to N_d depends on horizontal grid spacing between 40 km and 312 m , how much of it can be attributed to precipitation suppression and to droplet sedimentation, which boundary-layer processes accompany its change with grid spacing, and how close the simulations are to the in a single model, EAMxx, the atmosphere component of E3SM that implements SCREAM. We use a factorial set of doubly periodic simulations of the drizzling DYCOMS-II RF02 observations and published intercomparisons. Following Mülmenstädt et al. (2024a), case, in

which we scale the droplet fall speed ~~; in addition, we~~ (as in Mülmenstädt et al., 2024a) and switch off the N_d dependence of autoconversion. ~~This mechanism-denial approach allows the two pathways to be separated, subject to the caveats discussed in Sect. (prev.)~~The analysis is organised around one mixed-layer argument, in which grid spacing changes the entrainment rate and hence the depth, base and precipitation of the cloud, and the precipitation-to-entrainment balance of this base state sets the susceptibility. We test each link with diagnostics from the existing output, assess how much of the change in entrainment is resolved, parameterised or implicit, and quantify the transience and noise of the result. Because the case is nocturnal and the forcing is held constant for 12 h, all results describe a nocturnal adjustment that is still under way at the end of the simulations.

2 Model, case and methods

2.1 Model

EAMxx is the C++ atmosphere model of the Energy Exascale Earth System Model (E3SM) and implements SCREAM (Caldwell et al., 2021; Donahue et al., 2024). It solves the nonhydrostatic equations with a spectral-element dynamical core on fourth-order elements (Taylor et al., 2020). Physics is computed on a ~~separate~~ finite-volume grid with 2×2 columns per element ~~("pg2"; Hannah et al., 2021), so that~~ (Hannah et al., 2021), so physics columns are spaced 1.5 times the nominal grid spacing Δx, ~~defined as~~ the mean spacing of the dynamical-core quadrature points. Within each physics step the processes are called in the order SHOC, cloud fraction, prescribed aerosol, P3 and radiation. Turbulence, cloud macrophysics and condensation are represented by the Simplified Higher-Order Closure (SHOC; Bogenschutz and Krueger, 2013), an assumed-PDF scheme ~~that diagnoses cloud fraction and liquid water from~~ with a prognostic turbulent kinetic energy ~~. SHOC limits its turbulent mixing length to the host grid length, which in this configuration is the~~ whose mixing length is limited to the nominal Δx. Condensation and evaporation of cloud water are part of SHOC's saturation adjustment and do not depend on droplet size~~. We verified this~~ , which we verified in the output~~, where~~ : the vapour–liquid exchange of the microphysics ~~scheme~~ equals its rain evaporation at every level. The only routes by which N_d can affect entrainment in this model are therefore drizzle, droplet sedimentation and the effective radius used in radiation.

Microphysics is computed by the Predicted Particle Properties scheme (P3; Morrison and Milbrandt, 2015). Warm rain forms by the autoconversion and accretion rates of Khairoutdinov and Kogan (2000), without ~~the~~ a subgrid cloud-water variance enhancement of autoconversion~~used in some GCM schemes (Morrison and Gettelman, 2008; Gettelman and Morrison, 2015) . Rain fraction is set to one, meaning that~~ , and rain is assumed to fill the grid cell~~rather than to overlap with the liquid cloud fraction~~. Cloud droplets sediment with mass- and number-weighted fall speeds of a gamma size distribution. The P3 sedimentation-tendency diagnostics are not usable in ~~the model version used here,~~ this model version because the tendency buffer is not initialised~~with the pre-sedimentation mixing ratio; this affects only the diagnostics, not the model state, so the sedimentation flux at cloud top could not be diagnosed.~~, so the cloud-top sedimentation flux was estimated offline (Sect. 2.4). Droplet activation uses ~~a prescribed CCN number concentration from~~ the Simple Prescribed Aerosol option (SPA; Mahfouz et al., 2025)~~. In each layer,~~ : at the start of P3 ~~raises~~ in each step, the in-cloud droplet number is raised to at least the prescribed CCN number multiplied by a scaling factor. Because the prescribed CCN number decreases with height, this floor falls from

about 345 to 150 $\mathrm{cm^{-3}}$ between 300 and 900 m for a target of 200 $\mathrm{cm^{-3}}$ (Appendix B). Radiation is computed by RRTMGP (Pincus et al., 2019) every third physics step.

We used the doubly periodic configuration ~~of EAMxx (DP-EAMxx, also referred to as DP-SCREAM; Bogenschutz et al., 2023a, 2025) , which applies large-scale forcing from an intensive-observation-period (IOP) file to a horizontally periodic domain. All simulations used~~ (DP-EAMxx or DP-SCREAM; Bogenschutz et al., 2023a, 2025) with 128 vertical levels ~~, with a spacing of about~~ and a level spacing of 50 m ~~near the inversion. An optional horizontal ("3D") turbulence closure, which applies SHOC's eddy diffusivity in the horizontal, was off in the main experiment and was tested at 625~~between 500 and 1200 ~~m and 312~~m. All runs use the same random seed for the initial temperature perturbation (a relative perturbation of at most 0.1 ~~m (Appendix C).~~ % below 900 hPa), so runs that differ in N_d start from identical states. Two runtime options were added to P3~~for this study. The first is a~~. A sedimentation factor s ~~that multiplies the mass- and number-weighted~~ multiplies the cloud-droplet fall speeds~~, analogous to the `cld_sed` parameter of the MG2 scheme (Morrison and Gettelman, 2008; Gettelman and Morrison, 2015) and to the sedimentation experiments of Mülmenstädt et al. (2024a)~~; $s = 1$ is the default. The ~~second makes the prefactor and N_d exponent of the autoconversion rate configurable,~~ autoconversion rate is

$$\left(\frac{\partial q_\mathrm{r}}{\partial t}\right)_\mathrm{auto} = A\, q_\mathrm{c}^{2.47} \left(\frac{N_\mathrm{d}}{1\,\mathrm{cm^{-3}}}\right)^{-\beta}, \tag{1}$$

~~where~~ with q_c ~~is~~ the in-cloud cloud water mixing ratio ($\mathrm{kg\,kg^{-1}}$) and the rate ~~is~~ in $\mathrm{kg\,kg^{-1}\,s^{-1}}$. The default~~values~~ , $A = 1350$ and $\beta = 1.79$~~define the configuration with precipitation suppression on. With suppression off~~, is referred to as N_d-dependent autoconversion. In the N_d-independent configuration, $\beta = 0$ and $A = 1350 \times 63^{-1.79} = 0.812$, so ~~that~~ the two rates are equal at ~~a diagnosed N_d of 63 $\mathrm{cm^{-3}}$~~ $N_\mathrm{d} = 63\,\mathrm{cm^{-3}}$, close to the ~~observed~~ RF02 value of ~~about~~ 55 $\mathrm{cm^{-3}}$~~used in the intercomparisons (Ackerman et al., 2009; Wyant et al., 2007)~~. The ratio of the ~~off to the on autoconversion rate at fixed q_c is~~ two rates, $(N_\mathrm{d}/63)^{1.79}$, ~~which~~ is 0.13 at 20 $\mathrm{cm^{-3}}$ and about 140 at 1000 $\mathrm{cm^{-3}}$~~. Switching suppression off therefore removes the direct~~ , so the N_d~~control of rain formation but also changes the amount of rain substantially, in opposite directions at low and~~ -independent runs drizzle heavily at high N_d (0.83 $\mathrm{mm\,d^{-1}}$ at 40 km and target 1000 $\mathrm{cm^{-3}}$).

2.2 Case

DYCOMS-II RF02 sampled a nocturnal, heavily drizzling stratocumulus deck over the north-eastern Pacific (Stevens et al., 2003; vanZanten et al., 2005; vanZanten and Stevens, 2005). ~~The case was~~ It is the basis of an LES intercomparison (Ackerman et al., 2009) and a single-column model intercomparison (Wyant et al., 2007). ~~The observations~~ Observations give a mean LWP of about $100 \pm 20\,\mathrm{g\,m^{-2}}$, an entrainment rate of 6.0–7.6 $\mathrm{mm\,s^{-1}}$ (Faloona et al., 2005), a ~~mean~~ surface drizzle rate of 0.35 $\mathrm{mm\,d^{-1}}$ and an evaporated fraction of ~~about~~ 73 % of the cloud-base precipitation (vanZanten et al., 2005; Wyant et al., 2007). ~~We used the E3SM single-column IOP file for this case.~~ The initial boundary layer has $\theta_l = 288.3$ K and $q_t = 9.45\,\mathrm{g\,kg^{-1}}$ below an inversion at 795 m, ~~which gives a lifting condensation level of about 400~~above which θ_l jumps to 295 ~~m. Surface~~ K. Following the intercomparison protocol, the surface sensible and latent heat fluxes ~~were~~ are prescribed as 16 and 93 $\mathrm{W\,m^{-2}}$ ~~,~~ and large-scale subsidence ~~was applied with~~ has a divergence of $3.75 \times 10^{-6}\,\mathrm{s^{-1}}$~~, and~~ . Shortwave radiation is off.

Three aspects of our configuration differ from the protocol. Radiation is computed by RRTMGP rather than by the parametric longwave scheme of the intercomparison, which included a term that offsets subsidence warming above the inversion; here temperature and moisture ~~were nudged towards the initial profiles~~ are nudged only above 650 hPa~~with a time scale of 1~~. In practice θ_l at 2 ~~h. Shortwave radiation was off.~~

~~The configuration departs from the intercomparison protocol in three respects. Winds were~~ km changes by only -0.3 K over 12 h and θ_l 300 m above the inversion by at most 0.4 K, identically at all grid spacings, because clear-sky cooling ($-2.6\,\mathrm{K\,d^{-1}}$) nearly balances subsidence warming. Winds are not nudged and there ~~was~~ is no Coriolis force~~, so the initial wind shear decayed freely. The forcing was~~ ; the mixed-layer wind decays from 8.4 to 5.8–6.1 $\mathrm{m\,s^{-1}}$ and the wind difference across the inversion is 3.8–4.6 $\mathrm{m\,s^{-1}}$ at all grid spacings, so shear differs little between runs. Droplet number varies with height as described above, instead of being constant at 55 $\mathrm{cm^{-3}}$. The forcing is held constant for 12 h, twice the ~~length of the protocol , to examine the evolution of the adjustment. The surface fluxes were prescribed, which removes the feedback of boundary-layer warming and drying on the fluxes; we estimate this feedback in Sect. 4.3. Unless stated otherwise we analyse~~ protocol length, so hours 10–12 ~~, and Sect. (prev.) uses hours 4–6 for comparison with the intercomparisons.~~ follow ten hours of night, have no counterpart in nature and correspond to a continued nocturnal adjustment.

2.3 Experiment design

Table 1 lists the factors and Table 2 the numerical settings. The main matrix crosses eight grid spacings, six target droplet numbers, three sedimentation factors and ~~precipitation suppression on or off, which gives~~ both autoconversion settings, giving 288 simulations~~with 3D turbulence off~~. A further 72 simulations repeated the 625 m and 312 m grids with a horizontal (“3D~~turbulence on; one of them, at 312 m, target $N_\mathrm{d} = 1000\,\mathrm{cm^{-3}}$, $s = 2$ with suppression on, did not complete and is excluded, so~~ ”) turbulence option; 71 ~~enter Appendix C.~~

~~Table 2 gives the numerical settings. The domain is 240~~completed (36 at 625 ~~km wide for $\Delta x \le 10$ km, and 600~~ m and ~~300~~35 at 312 ~~km at 40 and 20 km so that the coarse domains still contain 10×10 physics columns~~m), and they serve here as perturbed twins. The physics time step decreases from 100 s at $\Delta x \ge 5$ km to 15 s at 312 m, and the radiation interval ~~correspondingly~~ from 300 to 45 s~~. The ratio of physics to dynamics time step is 12 for $\Delta x \ge 2.5$ km and 24 for $\Delta x \le 1.25$ km, chosen for stability of the dynamical core. Time step and radiation interval therefore co-vary with Δx, and their effects cannot be separated from those of grid spacing in this experiment. The 625 m and 312 m runs were integrated in restart segments, which we verified to be bit-for-bit identical to continuous integrations.~~

~~The in-cloud N_d produced by a given CCN scaling factor depends on the height of the cloud, because the prescribed CCN number decreases with height. For a target of 200 $\mathrm{cm^{-3}}$, the activation floor decreases from about 345 $\mathrm{cm^{-3}}$ below 300 m to about 150 $\mathrm{cm^{-3}}$ near 900 m (Appendix B, Fig. B1b). At $\Delta x \ge 10$ km, where the cloud extends down to about 200 m, the diagnosed in-cloud N_d is 21–64 % above target, partly because small SHOC cloud fractions near cloud base inflate the in-cloud value. At 312 m, where the cloud lies between about 650~~ , so time step and ~~1000 m, it is between 21 % below and 11 % above target. The diagnosed in-cloud N_d, defined as the median over cloudy grid cells (cloud water above $10^{-5}\,\mathrm{kg\,kg^{-1}}$) of the 3D~~

Table 1. Experiment factors. Every combination was run for 12 h with 3D turbulence off (288 runs). The 625 m and 312 m grids were repeated with 3D turbulence on (72 runs, 71 completed).

Factor	Levels	Implementation
Grid spacing Δx	40, 20, 10, 5, 2.5, 1.25 km; 625, 312 m	Table 2
Target N_d	20, 50, 100, 200, 500, 1000 cm^{-3}	SPA CCN scaling factor (exponent 1)
Sedimentation factor s	0.5, 1, 2	multiplies cloud-droplet fall speeds in P3
Autoconversion	N_d-dependent, N_d-independent	Eq. (1) with $(A, \beta) = (1350, 1.79)$ or $(0.812, 0)$
3D turbulence	off; on at 625 and 312 m	SHOC eddy diffusivity applied horizontally

Table 2. Numerical settings. Δx is the nominal (dynamics) grid spacing; physics columns are $1.5\Delta x$ apart. N_col is the number of physics columns, Δt_phys and Δt_dyn the physics and dynamics time steps, and ν_top the sponge-layer diffusivity.

Δx	domain (km)	elements	N_col	Δt_phys (s)	Δt_dyn (s)	ν_top ($\mathrm{m}^2\,\mathrm{s}^{-1}$)
40 km	600	5×5	100	100	8.33	1.2×10^5
20 km	300	5×5	100	100	8.33	6.0×10^4
10 km	240	8×8	256	100	8.33	3.0×10^4
5 km	240	16×16	1024	100	8.33	1.5×10^4
2.5 km	240	32×32	4096	75	6.25	7500
1.25 km	240	64×64	16 384	60	2.50	1667
625 m	240	128×128	65 536	30	1.25	833
312 m	240	256×256	262 144	15	0.625	417

All runs used 128 levels, radiation every third physics step and one macrophysics–microphysics subcycle per physics step.

~~snapshots at 10 and 12 h, changes by less than 17 % between 6 and 12 h at $\Delta x \leq 5$ km and by up to 25 % at $\Delta x \geq 10$ km.~~ grid spacing co-vary.

2.4 Diagnostics and notation

~~Diagnostics were computed from domain-mean profiles averaged over 15 min, from instantaneous column fields every 15 min, from 2-hourly 3D snapshots, and from cloud-top fields every 15 min. The cloud-top fields were produced by an output operator that follows the AeroCom convention of reporting the liquid cloud-top fraction and the fraction-weighted droplet number and effective radius of the uppermost liquid layer.~~ LWP is the domain mean averaged over hours 10–12 unless stated otherwise. Cloud cover ~~is reported both as the fraction of columns with LWP above 10 $\mathrm{g\,m^{-2}}$, and as the~~ f_c is the maximum-overlap ~~cloud~~ cover computed from SHOC's liquid cloud fraction in the 3D snapshots ~~, which includes subgrid cloud.~~ at 10 and 12 h; the fraction of columns with LWP above 10 $\mathrm{g\,m^{-2}}$ is used as an alternative. In-cloud LWP is $\mathrm{LWP}_\mathrm{ic} = \mathrm{LWP}/f_\mathrm{c}$. The inversion height z_i is the ~~height at which the domain-mean liquid potential temperature~~ midpoint of the θ_l ~~crosses the midpoint between~~

~~its boundary-layer (below 400 m) and free-tropospheric (1400 m) values. Cloud~~ jump in the domain-mean profile, and cloud base z_b ~~is~~ the lowest level at which the domain-mean liquid cloud fraction reaches half its maximum~~; it is quantised by the 25–50 m level spacing. The entrainment rate is~~ . The quantity $w_e = dz_i/dt - w_s(z_i)$, ~~where~~ with dz_i/dt fitted over hours 8–12 and w_s ~~is the IOP subsidence velocity~~ the prescribed subsidence (negative downward)~~at the mean z_i and dz_i/dt is a linear fit over hours 8–12. The 4 h fit is less sensitive than a 2 h fit to the 50 m level spacing at the inversion, across which z_i rises by about 70 m at 40 km and about 200 m at 312 m over 12 h. Vertical-velocity variance and skewness of the resolved flow were computed on the physics grid from~~ , is the ~~3D snapshots at 10 and 12~~subsidence-corrected deepening rate; we call it the entrainment rate for brevity, although it includes any implicit numerical mixing at the inversion. Cloud-top N_d is the fraction-weighted droplet number of the uppermost liquid layer from an output operator that follows the AeroCom convention, divided by the cloud-top liquid fraction. Unlike the in-cloud layer median, it varies little between grid spacings at fixed target (151–162 ~~h, with $w = -\omega/(\rho g)$, and averaged over the boundary layer. They are compared with SHOC's subgrid variance, which SHOC diagnoses from its turbulent kinetic energy and is therefore not an independent second moment. Decoupling is measured as the difference in q_t between the lowest and highest quarters of the boundary layer (Jones et al., 2011). The spectral peak wavelength λ of LWP is the maximum of the radially averaged power spectrum (Wood and Hartmann, 2006); with a 240 km domain it is quantised at $240/n$ km~~cm^{-3} at target 200), so we use it as the abscissa of all susceptibilities. It is sampled at the end of the time step, after P3 processes have depleted droplets below the activation floor.

We define the LWP susceptibility between ~~two otherwise identical~~ runs at target droplet numbers a and b as

$$S_{a\to b} = \frac{\ln \mathrm{LWP}_b - \ln \mathrm{LWP}_a}{\ln N_{\mathrm{d},b} - \ln N_{\mathrm{d},a}},$$

~~where LWP is averaged over hours 10–12 and N_d is the diagnosed in-cloud value. The subscripts refer to the target N_d, while the slope uses the diagnosed N_d. We report three~~ $S_{a\to b} = \Delta \ln \mathrm{LWP}/\Delta \ln N_d$ and report four branches, $S_{20\to50}$, $S_{50\to100}$, $S_{100\to200}$ and $S_{200\to1000}$, ~~and denote susceptibilities with precipitation suppression on and off by S_{on} and S_{off}. Because~~ labelled by target N_d ~~is diagnosed from snapshots while LWP varies continuously, we recomputed~~ but computed with cloud-top N_d. Because $\mathrm{LWP} = f_c \mathrm{LWP}_{ic}$, $S = S_{ic} + S_c$ with $S_{ic} = \Delta \ln \mathrm{LWP}_{ic}/\Delta \ln N_d$ and $S_c = \Delta \ln f_c/\Delta \ln N_d$. We use "susceptibility" for S and reserve "adjustment" for its radiative meaning. For adiabatic clouds the optical-depth susceptibility is $d\ln\tau/d\ln N_d = 1/3 + (5/6$ To relate S ~~for each of the eight 15 min intervals in hours 10–12 with fixed~~ to albedo we computed an offline scene albedo $f_c\tau/(\tau+13.3)$ with $\tau = 0.07\,\mathrm{LWP}_{ic}^{5/6} N_d^{1/3}$ and split its change between two N_d ~~and report the standard deviation. These 15 min samples are autocorrelated and come from a single realisation, so this spread measures temporal sampling only and is not an estimate of the uncertainty that an initial-condition ensemble would provide. Comparisons across grid spacings at fixed N_d use~~ levels sequentially into Twomey, in-cloud LWP and ~~other quantities interpolated linearly in $\ln N_d$ to fixed diagnosed values, because the diagnosed~~ cloud-cover terms (Platnick and Twomey, 1994). Because the case is nocturnal, these are potential albedo responses, not forcings. The precipitation susceptibility is $S_R = -\Delta \ln P/\Delta \ln N_d$ (Terai et al., 2012).

The base-state regime of a pair of runs is characterised by the ratio of surface precipitation to entrainment drying, $P/F_{ent,q}$ with $F_{ent,q} = -\rho L_v w_e \Delta q_t$, of the run at the lower N_d~~at a given target differs between grid spacings.~~ , the diagnostic that Ackerman et al. (2004) used to explain the sign of the LWP response. Two entrainment efficiencies are used. The first, $A = w_e z_i \Delta b/w_*^3$

with $w_*^3 = 2.5\,(g/\theta_0)\int_0^{z_i} \overline{w'\theta_v'}\,dz$ from the resolved plus SHOC buoyancy flux and Δb from the θ_l and q_t jumps (Stevens, 2002), does not contain the radiative forcing. The second, $\eta = \rho c_p w_e \Delta\theta_l / F_{\mathrm{rad}}$, compares entrainment warming with the radiative cooling F_{rad} of the boundary layer; its value depends on how the jump is defined, so we report it for $\Delta\theta_l$ taken 50, 100 and 150 m above z_i and for the specified interface jump of 6.7 K. In a steady state $\eta = 1 - (H + P)/F_{\mathrm{rad}}$, which is 0.7–0.8 at $N_d \geq 200\,\mathrm{cm}^{-3}$.

~~For each grid spacing and branch, we fitted $S_{\mathrm{on}}(s) = a_{\mathrm{on}} + b_{\mathrm{on}} s$ and $S_{\mathrm{off}}(s) = a_{\mathrm{off}} + b_{\mathrm{off}} s$ over $s = 0.5$, 1 and 2, and decomposed the susceptibility at $s = 1$ as~~

$$S_{\mathrm{on}}(1) = (a_{\mathrm{on}} - a_{\mathrm{off}}) + b_{\mathrm{off}} + (b_{\mathrm{on}} - b_{\mathrm{off}}) + a_{\mathrm{off}}.$$

~~The first term is the precipitation-suppression contribution extrapolated to zero sedimentation, the second the sedimentation contribution in the absence of precipitation suppression, the third the interaction between sedimentation and precipitation suppression, and the fourth a residual that includes radiative effects of the effective radius and any non-linearity in s. The sum reproduces $S_{\mathrm{on}}(1)$ to within the fit error. Three caveats apply. The extrapolation to $s = 0$ uses three points with one residual degree of freedom, and S_{off} is clearly convex in s on the clean branch at sub-kilometre spacing, so a_{on} and a_{off} are uncertain there. On and off runs at the same target have slightly different diagnosed N_d, so their slopes span different $\ln N_d$ intervals. Most importantly, the off configuration is matched at 63~~From the 3D snapshots at 10 and 12 ~~cm^{-3} and therefore precipitates heavily at high N_d; at 40~~h we computed the resolved turbulent θ_l flux $\overline{w'\theta_l'}$ and its minimum within 150 ~~km and target 1000~~m below z_i, the SHOC flux at that level, the inversion thickness (the depth over which $d\theta_l/dz$ exceeds 20 ~~cm^{-3}, surface precipitation is 0.83~~% of its maximum), the spread of column cloud-top height, the share of resolved vertical-velocity variance at mid-boundary-layer wavelengths shorter than six physics-column spacings, and the autoconversion enhancement $\langle q_c^{2.47}\rangle / \langle q_c\rangle^{2.47}$ by resolved variability. An offline estimate of the cloud-top sedimentation flux uses the in-cloud cloud water and droplet number at the highest cloudy level of each column and the P3 gamma distribution (shape parameter from N_d, Stokes fall speeds), $F_{\mathrm{sed}} = s\rho q_c V_m$ with $V_m = a_c(\mu+5)(\mu+4)/\lambda^2$. Decoupling is described by four metrics: the two-part test of Jones et al. (2011) ($\Delta q_t > 0.5$ ~~mm~~g ~~d~~kg^{-1} ~~with suppression off and 0.01~~and $\Delta\theta_l > 0.5$ ~~mm~~K between the lowest and highest quarters of the layer below the inversion base), the height of cloud base above the lifting condensation level of mixed-layer air, the minimum of the total buoyancy flux below cloud base relative to its surface value (Turton and Nicholls, 1987; Bretherton and Wyant, 1997), and the ratio of cloud-layer to subcloud vertical-velocity variance.

Uncertainty of S from internal variability was estimated in two ways. The 3D-turbulence runs differ from their partners by a small perturbation of the subgrid mixing, so their differences in S estimate run-to-run noise; their mean LWP difference is 0.4 ~~d^{-1} with suppression on. The off runs at coarse spacing thus measure the sensitivity to sedimentation in a strongly precipitating state, whereas at 312~~% and the root-mean-square difference 0.6 ~~m the off runs precipitate only 0.01–0.17~~%. Spatial uncertainty was estimated by bootstrapping 16 tiles of 60 ~~mm~~km over the eight 15 ~~d^{-1}. The partition between the first two terms at coarse spacing therefore depends on the choice of the matching N_d and should be read as specific to this design.~~ min instants in hours 10–12. Both estimates share the initial perturbation and do not sample structural uncertainty.

2.5 Mixed-layer framework

To relate the LWP response to boundary-layer processes, we evaluated a bulk budget of the domain-mean boundary layer for every run in 2 h windows, following Following mixed-layer theory (Lilly, 1968; Stevens, 2002) and the LWP-tendency framework of van der Dussen et al. (2014). For a well-mixed layer of depth z_i with conserved variables , the boundary-layer means of θ_l and q_t , evolve as

$$\frac{d\theta_l}{dt} = \frac{1}{z_\mathrm{i}}\left[\frac{H}{\rho c_p} + w_\mathrm{e}\,\Delta\theta_l - \frac{F_\mathrm{rad}}{\rho c_p} + \frac{P}{\rho c_p}\right], \tag{2}$$

$$\frac{dq_t}{dt} = \frac{1}{z_\mathrm{i}}\left[\frac{E}{\rho L_v} + w_\mathrm{e}\,\Delta q_t - \frac{P}{\rho L_v}\right], \frac{dz_\mathrm{i}}{dt} = w_\mathrm{e} + w_\mathrm{s}(z_\mathrm{i}), \tag{3}$$

where H and E are the surface sensible and latent heat fluxes, F_rad is the radiative cooling of the layer below $z_\mathrm{i} + 150$ m, P is the surface rain rate in energy units, and $\Delta\theta_l$ and Δq_t are the differences between the values 100 m above z_i and the mixed-layer means over 0.1–0.8 $z_\mathrm{i}$$dz_\mathrm{i}/dt = w_\mathrm{e} + w_\mathrm{s}(z_\mathrm{i})$. Cloud base was taken as follows the lifting condensation level of mixed-layer air, so that , $dz_\mathrm{b}/dt = (\partial z_\mathrm{b}/\partial\theta_l)\,d\theta_l/dt + (\partial z_\mathrm{b}/\partial q_t)\,dq_t/dt$, with sensitivities of 120–130evaluated for cloud-layer air (0.6–0.9 m K^{-1} and -190 to -210 m (g kg^{-1})$^{-1}$ computed with the saturation formula of Bolton (1980). Assuming LWP $\propto h^2$ with cloud thickness z_i). Assuming $\mathrm{LWP_{ic}} \propto h^2$ with $h = z_\mathrm{i} - z_\mathrm{b}$ and fixed adiabatic fraction and cloud cover, ,

$$\frac{d\ln \mathrm{LWP}}{dt} = \frac{2}{h}\left(\frac{dz_\mathrm{i}}{dt} - \frac{dz_\mathrm{b}}{dt}\right) + \frac{d\ln f_\mathrm{c}}{dt}, \tag{4}$$

which separates into contributions from entrainment (through where the first term is evaluated from the forcing terms and the cloud-cover term is taken from the simulations. The entrainment contribution splits into a deepening part ($2w_\mathrm{e}/h$) and a cloud-base part (the effect of w_e in dz_i/dt and in Eqs. 2 and 3), subsidence, radiation, surface fluxes and precipitation.

on θ_l and q_t). Because w_e was is diagnosed from the simulated dz_i/dt, the budget 's dz_i/dt equals the simulated value by construction, and only the cloud-base tendency and the LWP–thickness relation are predicted. The budget should therefore be read as a diagnostic decomposition. It also assumes a well-mixed layer, whereas the low-N_d boundary layers at $\Delta x \leq 5$is a diagnostic decomposition and not an independent closure, and it is meaningful only where the layer is coupled. We therefore apply process attribution only to 2 km are decoupled (Sect. (prev.)) and the sub-kilometre runs have cloud cover well below one. Pooled over all 1440 run-windows with 3D turbulence off, the predicted and simulated tendencies are correlated at 0.89 for θ_l, 0.87 for q_t and 0.88 for LWP. The skill varies strongly with grid spacing (Appendix (prev.), Table (prev.)). For LWP it is high at $\Delta x \geq 5$h windows that pass the Jones test, have cloud base within 50 km (correlation 0.79–0.90) and low at sub-kilometre spacing (0.07 at 625 m m of the mixed-layer lifting condensation level, and 0.44 at 312have a minimum buoyancy flux below cloud base of more than 20 m, with errors comparable to the signal). For the differences between N_d levels in hours 10–12, which determine S, the correlation is 0.85–0.93 at 40, 20 and 5 km and -0.02 to 0.57 at $\Delta x \leq 2.5$ km% of the surface value.

3 Results

3.1 Three regimes of the mean state

Table 3 compares hours 4–6 of the runs at target $N_d = 50\,\mathrm{cm^{-3}}$ (diagnosed 42–67The mean state falls into three regimes (Figs. 1 and 2). At $\Delta x \geq 10\,\mathrm{cm^{-3}}$) and $s = 1$ with the observations and with the intercomparisons, which averaged over the last two of their 6km all vertical-velocity variance is parameterised, the columns are nearly identical, and the inversion is pinned near a model interface. At 40 h. The runs at $\Delta x \geq 5$ km have an LWP of 156–159z_i rises by only 45 $\mathrm{g\,m^{-2}}$, above the observed $100 \pm 20\,\mathrm{g\,m^{-2}}$ and similar to the DHARMA LES value of 144m over 12 $\mathrm{g\,m^{-2}}$ reported by Wyant et al. (2007) . Their entrainment rate of 4.1h, w_e is 3.1–4.5 mm $\mathrm{s^{-1}}$ is below the observed 6.0–7.6^{-1}, barely more than the subsidence velocity of about 3.2 mm $\mathrm{s^{-1}}$ and close to the LES value of 4.5, and at 40 km and target $N_d \geq 100\,\mathrm{cm^{-3}}$ it varies only between 3.6 and 3.9 mm $\mathrm{s^{-1}}$, across all sedimentation and autoconversion settings. The cloud extends down to 200–250 m and their surface precipitation of 0.12–0.14LWP reaches 300–340 mmg $\mathrm{m^{-2}}$ at high N_d. At 5 $\mathrm{d^{-1}}$ is below the observed 0.35km the run follows the coarse solution for about 8 mmh ($z_i = 862$ $\mathrm{d^{-1}}$. With decreasing grid spacing, LWP decreases to 98m at 8 $\mathrm{g\,m^{-2}}$ h compared with 857 m at 312 m and the entrainment rate increases to 40 km) and then deepens rapidly, with w_e rising to about 7.7 mm $\mathrm{s^{-1}}$, at the upper end of the observed range. Cloud base rises from about 300in hours 8–12. The resolved flow at 5 m to 580km carries only 3–7 m, compared with the lifting condensation level of about 400% of the vertical-velocity variance, and 60–74 m of the initial mixed layer. The 625% of that variance lies at wavelengths shorter than six physics-column spacings, in the stripe-like structures visible in Fig. 1; the 5 m runs are closest to the observations overall, with an LWP of 119km runs are therefore a transitional, partly numerical state. At $\Delta x \leq 2.5$ $\mathrm{g\,m^{-2}}$, an entrainment rate of 6.3km the resolved share of the variance rises to 54–67 mm% at 312 $\mathrm{s^{-1}}$, surface precipitation of 0.28m and the near-grid share falls to 28–31 mm%, at target 200 $\mathrm{d^{-1}}$ and an evaporated fraction of 0.73, equal to the observed value. No single grid spacing matches all observed quantities, and LWP, entrainment and precipitation at all spacingslie within the spread of the single-column models, whose LWP ranged from 30 $\mathrm{cm^{-3}}$ the cloud base rises from about 490 to 162720 $\mathrm{g\,m^{-2}}$ and entrainment rate from 1.5 to 6.8m, the cloud thins from about 450 to 300 mmm, and LWP decreases by 21–40 $\mathrm{s^{-1}}$ (Wyant et al., 2007). Ackerman et al. (2009) found that their LES ensemble with drizzle and sedimentation had a steady LWP consistent with the measurements and an entrainment rate at the low end of the measured range, and that it underestimated precipitation. The coarse EAMxx runs share the low entrainment and low precipitation, while the sub-kilometre runs entrain more than the LES. We did not run an LES of the same configuration, so a converged reference with this model physics is not available% between 625 and 312 m at fixed N_d, without signs of convergence.

Figure 1 shows LWP at hour 12 for $s = 1$ with precipitation suppression on. As the grid is refined, the LWP spectral peak shortens from about 30 km at 2.5 km to 13–22 km at 625 and The increase of w_e with decreasing grid spacing is not simply an increase of resolved turbulence. The total vertical-velocity variance at 312 m , the domain-mean LWP decreases, and clear gaps appear. Interpolated to a diagnosed in-cloud N_d of 200(about 0.09 $\mathrm{cm^{-3}}$, LWP is 273$\mathrm{m^2\ g\,m^{-2}\ s^{-2}}$) is a quarter of that at 40 km, 204the convective velocity scale is similar at all spacings (0.9–1.3 $\mathrm{g\,m^{-2}}$ at 5m km and 73 $\mathrm{g\,m^{-2}}$ at 312 m $\mathrm{s^{-1}}$), and the maximum-overlap cloud cover decreases from 1.00 at $\Delta x \geq 2.5$efficiency A rises from 0.6–0.8 at $\Delta x \geq 10$ km to

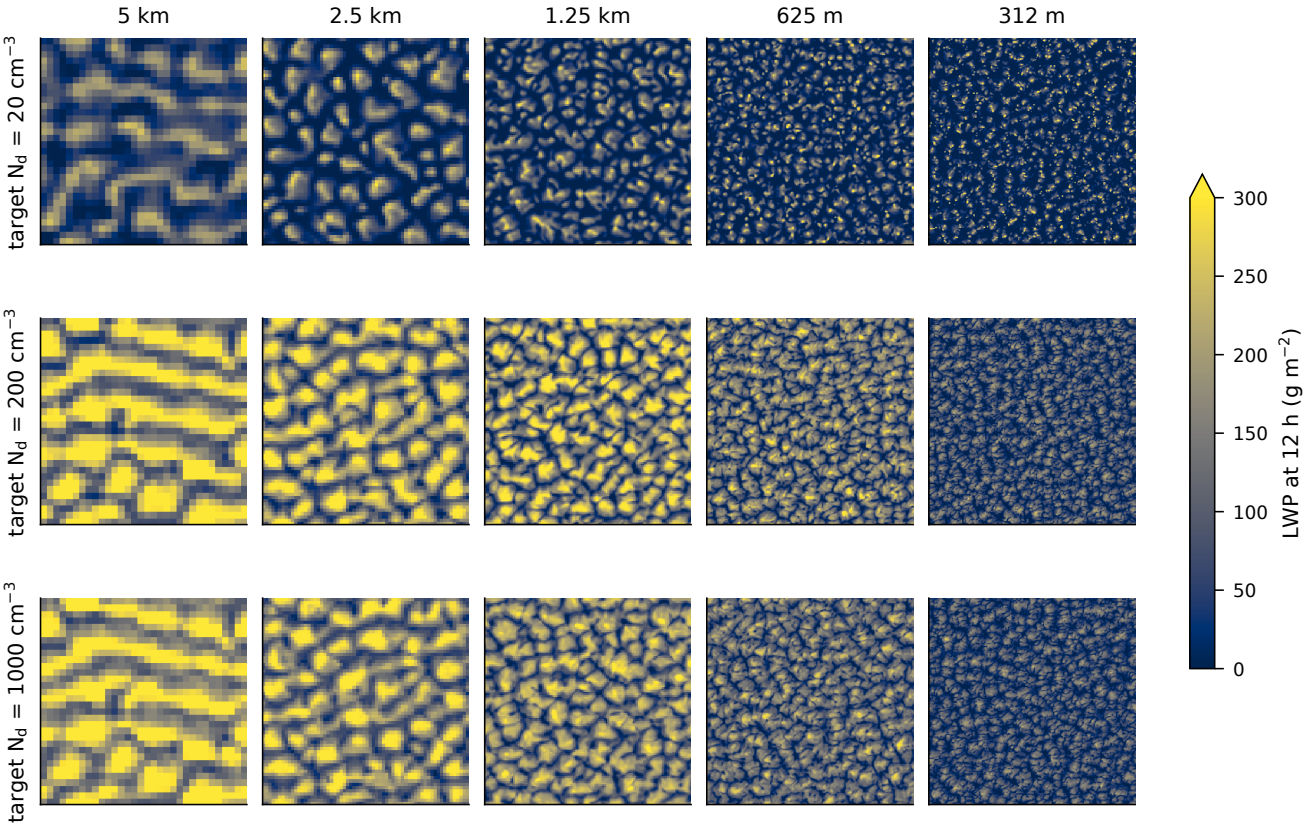


Figure 1. Instantaneous LWP at hour 12 over the 240 km domain for target N_d of 20, 200 and 1000 cm^{-3} (rows) and grid spacings from 5 km to 312 m (columns), for sedimentation factor 1 with N_d-dependent autoconversion.

~~0.92 at~~ 1.1–1.6 at 5–1.25 km, 1.5–1.8 at 625 m and 2.0–2.6 at 312 m ~~. At target $N_\mathrm{d} = 20\,\mathrm{cm}^{-3}$~~ (Fig. 2d). At $\Delta x \leq 5$ km and target $N_\mathrm{d} \geq 200\,\mathrm{cm}^{-3}$, the ~~sub-kilometre clouds break into small cells of high LWP, and the maximum-overlap cloud cover at 312~~resolved plus SHOC turbulent θ_l flux at its minimum below the inversion carries only 37–46 ~~m is 0.68 (0.50 with the LWP threshold). The 5~~% of the implied entrainment flux $-w_\mathrm{e}\Delta\theta_l$ (with the jump taken 50 ~~km fields contain stripe-like structures with a wavelength of about six physics columns, close to the effective resolution~~ m above z_i), a fraction that does not change systematically with grid spacing (Fig. 2f). The remainder, which includes radiative cooling within the entrainment zone and implicit mixing by the vertical remapping and advection of the dynamical core, ~~which suggests that the circulations at this spacingare partly numerical. Between 40~~is therefore as large as the explicit flux at every spacing. At target $N_\mathrm{d} \geq 200\,\mathrm{cm}^{-3}$ the inversion is two to three levels thick (100–175 ~~km and 312 m at target 200 cm^{-3}, the inversion rises from 863 to 1016~~ m~~, cloud base rises from 195 to 720~~) at all grid spacings, and the standard deviation of column cloud-top height at high N_d is 32–41 ~~m, the cloud layer thins from 668 to 296~~ m, ~~and the mixed layer becomes 1.8~~less than one level. The entrainment interface is thus not resolved vertically at any grid spacing, and SHOC's mixing length is at its cap of 312 ~~K warmer. The~~

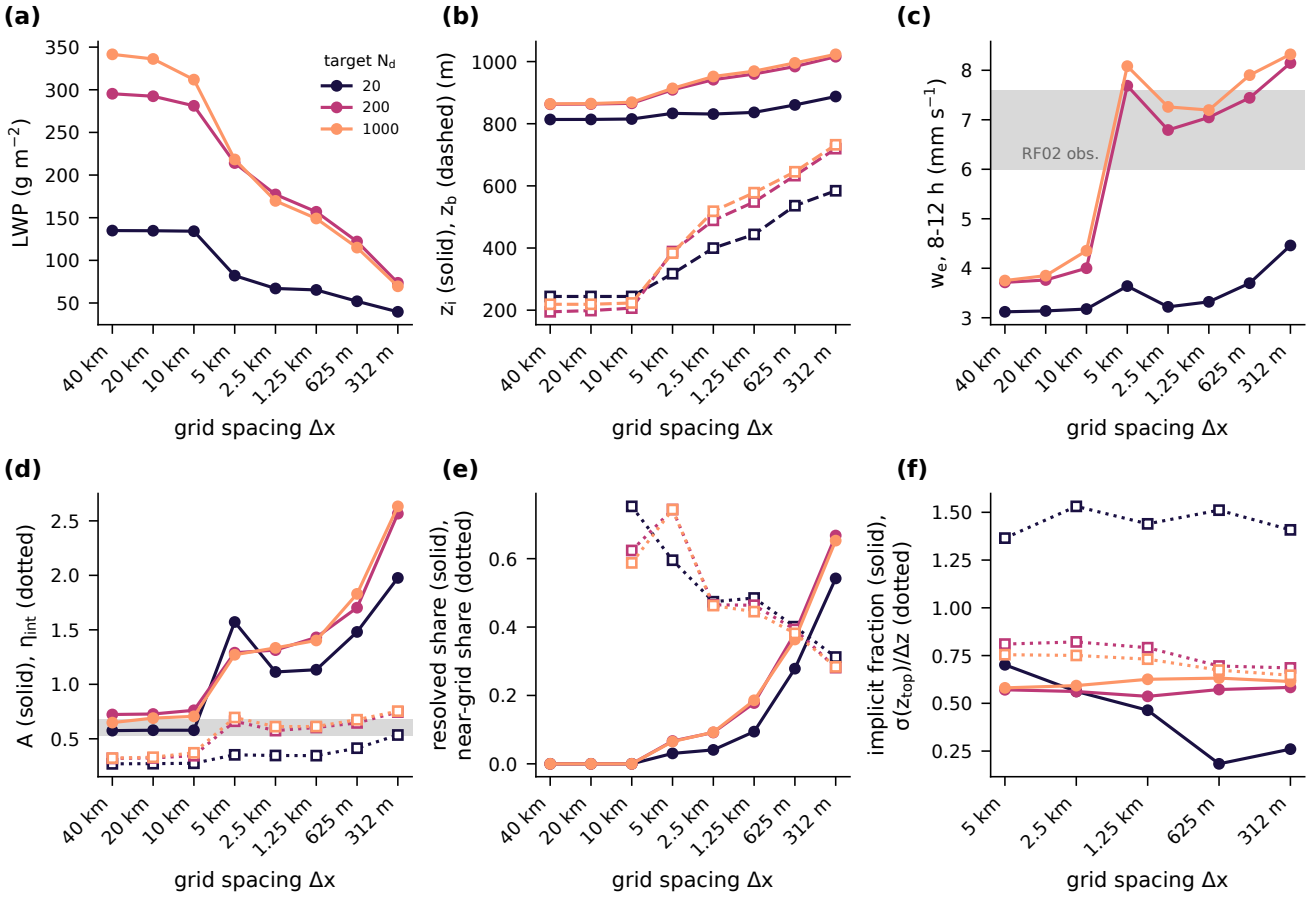


Figure 2. Mean state against grid spacing for target N_d of 20, 200 and 1000 cm^{-3} (sedimentation factor 1, N_d-dependent autoconversion, hours 10–12). (a) LWP. (b) Inversion height (solid) and cloud base (dashed). (c) Entrainment rate (8–12 h fit) with the observed range. (d) Entrainment efficiency A (solid) and η with the specified interface jump (dotted), with the observed η range. (e) Resolved share of boundary-layer vertical-velocity variance (solid) and share of resolved variance at wavelengths shorter than six physics-column spacings (dotted). (f) Fraction of the implied entrainment flux of θ_l not carried by resolved and SHOC fluxes (solid), and spread of column cloud-top height relative to the 50 m level spacing (dotted), for $\Delta x \leq 5$ km.

~~40 and 20~~m nearly everywhere in the boundary layer at 312 ~~km runs are nearly indistinguishable. The 10 km runs differ from them at high N_d (LWP~~ m. Together with the positive vertical-velocity skewness at 312~~instead of 342 $\mathrm{g\,m}^{-2}$~~ m (0.09–0.24 at target ~~1000~~$N_d \geq 100\,\mathrm{cm}^{-3}$), these diagnostics indicate that the 312 ~~cm^{-3}), where their column LWP standard deviation reaches up to 27 $\mathrm{g\,m}^{-2}$ across configurations, compared with at most 5 $\mathrm{g\,m}^{-2}$ at 20 and 40 km. Resolved motions carry no measurable vertical-velocity variance at $\Delta x \geq 10$ km (Sect. (prev.)), so these runs approximate a single-column configuration of the model physics, with the 10 km runs beginning to develop weak horizontal structure~~m runs are not an approximation to LES of RF02, which use spacings of a few metres near the inversion (Ackerman et al., 2009), and that part of the increase in

w_e with refinement may be numerical. Vertical refinement at the inversion would be needed to separate the physical from the numerical part (Stevens and Bretherton, 1999; Bogenschutz et al., 2023b).

~~Figure 3 and Table (prev.) show the LWP response to N_d.With precipitation suppression on, LWP increases monotonically with N_d~~ The efficiency η illustrates how strongly the interpretation depends on the jump definition (Fig. 8c). With the jump taken 100 m above z_i, η is 0.6–0.7 at $\Delta x \geq 10$ km ~~, from 135 to 342~~and 1.0–1.3 at finer spacing; with the specified interface jump of 6.7 ~~g m^{-2} at 40~~K it is 0.3–0.4 and 0.6–0.75, below the steady-state reference of 0.7–0.8. With the same definitions, the observed RF02 entrainment rate gives $\eta = 0.9$–1.2 and 0.53–0.68. The coarse runs therefore entrain too little compared with RF02, and the sub-kilometre runs entrain at the upper end of the observed range; $\eta > 1$ in the earlier definition reflected the 100 ~~km. With decreasing grid spacing, the curve flattens at high N_d and then turns over. The LWP maximum moves from target $N_d \geq 1000$ cm^{-3} at $\Delta x \geq 10$ km to 500 cm^{-3} at~~ m offset, which adds about 5 ~~km, 200 cm^{-3} from 2.5 km to 625 m, and 100 cm^{-3} (diagnosed 84 cm^{-3}) at 312 m. The resulting shape resembles the inverted-V shape in satellite composites and LES ensembles (Gryspeerdt et al., 2019; Glassmeier et al., 2021), but in these simulations it appears only at $\Delta x \leq 5$ km, and the negative slopes are an order of magnitude smaller than the observed ones, which Gryspeerdt et al. (2019) regarded as an upper bound on the causal effect~~K of the free-tropospheric θ_l profile.

~~With precipitation suppression off (Fig. 3b), LWP decreases slightly with N_d at $\Delta x \geq 10$~~Table 3 compares hours 4–6 at target $N_d = 50$ cm^{-3} with the observations and the published simulations. The runs at $\Delta x \geq 5$ km ~~(from 234 to 208~~resemble the DHARMA LES of Wyant et al. (2007) in LWP (156–159 compared with 144 g m^{-2}~~at 40~~) and entrainment (4.1 compared with 4.5 ~~km) and at 312~~mm ~~m (from 79 to 64~~s^{-1}) but precipitate less than observed. The 625 ~~g m^{-2}) . At 5~~m runs are closest to the observations, with an LWP of 119 ~~km it increases monotonically, from 135 to 160~~ g m^{-2}, ~~and at 2.5~~an entrainment rate of 6.3 ~~km to 625~~mm ~~m it increases between target 20 and 50~~s^{-1}, surface precipitation of 0.28 ~~cm^{-3} ($S_{20\to50} = 0.08$–0.20) and decreases slightly at higher N_d. Surface precipitation (Fig. 3d)decreases steeply with N_d when suppression is on, and more steeply at finer spacing; at 312 m it is 0.011~~ mm d$^{~~-1~~}$$^{-1}$ and an evaporated fraction of 0.73, although the evaporated fraction may be favoured by the assumption that rain fills the grid cell. No grid spacing matches all observed quantities, and by hours 10–12 the sub-kilometre runs have thinned further (LWP 70–120 g m^{-2} at target ~~100~~50–200 ~~cm^{-3} and~~ cm^{-3}).

3.2 Susceptibility after 10–12 h

With N_d-dependent autoconversion, LWP increases monotonically with N_d at ~~most 0.002~~$\Delta x \geq 10$ ~~mm d^{-1} at higher~~ km, whereas at finer spacing it flattens and then decreases slightly at high N_d ~~. The entrainment rate~~ (Fig. 3~~c)is 3.1–4.5 mm s^{-1} at $\Delta x \geq 10$ km and increases with~~). With N_d~~at $\Delta x \leq 5$ km whether suppression is on or off. It is not monotonic in Δx: at target $N_d \geq 100$ cm^{-3} it is larger at 5 km (7.0–8.1 mm s^{-1}) than at 2.5 km (5.7–7.3 mm s^{-1}), and it increases again towards 312 m (8.0–8.3 mm s^{-1}).~~

~~Figure (prev.) and~~ -independent autoconversion it is nearly flat at all spacings. Table 4 ~~give the corresponding susceptibilities. With suppression on and $s = 1$,~~ and Fig. 4 give the susceptibilities. The moderate branch $S_{100\to200}$ decreases ~~monotonically as Δx decreases, from 0.27~~ from 0.25 at 40 km to ~~0.20~~ 0.21 at 5 km, 0.11 at 2.5 km, 0.07 at 1.25 km, 0.03 at 625 m and -0.05 at 312 m, and it is entirely an in-cloud response. The polluted branch $S_{200\to1000}$ changes sign between 5 ~~km (0.01)~~ and

Table 3. Hours 4–6 at target $N_\mathrm{d} = 50\,\mathrm{cm}^{-3}$ and sedimentation factor 1, compared with RF02 observations and published simulations. P_sfc is surface precipitation and f_evap the fraction of rain production that evaporates before reaching the surface. Observations: LWP and f_evap as compiled by Wyant et al. (2007), w_e from Faloona et al. (2005), P_sfc from vanZanten et al. (2005). The LES entry is the single DHARMA simulation and the SCM range covers nine models, both averaged over hours 4–6 from Table 2 of Wyant et al. (2007) (standard vertical resolution).

Δx	autoconv.	LWP ($\mathrm{g\,m^{-2}}$)	w_e ($\mathrm{mm\,s^{-1}}$)	P_sfc ($\mathrm{mm\,d^{-1}}$)	f_evap	z_b (m)	z_i at 6 h (m)
40 km	on	159	4.2	0.14	0.85	303	849
20 km	on	158	4.1	0.13	0.85	303	849
10 km	on	157	4.1	0.12	0.86	299	850
5 km	on	156	4.1	0.13	0.85	303	851
2.5 km	on	132	4.5	0.68	0.52	432	864
1.25 km	on	132	5.1	0.52	0.61	443	870
625 m	on	119	6.3	0.28	0.73	506	888
312 m	on	98	7.7	0.10	0.84	578	913
40 km	off	171	4.2	0.05	0.91	303	851
20 km	off	171	4.2	0.05	0.91	303	852
10 km	off	170	4.2	0.05	0.92	313	852
5 km	off	168	4.2	0.06	0.90	313	853
2.5 km	off	141	4.5	0.52	0.55	432	866
1.25 km	off	142	5.3	0.42	0.63	443	872
625 m	off	125	6.8	0.23	0.73	501	891
312 m	off	103	7.7	0.08	0.85	572	914
Observations		100 ± 20	6.0–7.6	0.35	0.73		
DHARMA LES		144	4.5				
SCMs (9 models)		30–162	1.5–6.8				

2.5 km ~~(−0.03) and lies between −0.03~~ and is −0.03 to −0.04 at finer spacing~~.~~; it is negative in 23 of the 24 configurations (sedimentation factor and autoconversion setting) at $\Delta x \le 2.5$ km and becomes more negative with s there (−0.01 to −0.02 for $s = 0.5$ and −0.06 to −0.09 for $s = 2$).

The clean branch behaves differently. $S_{20\to50}$ increases from ~~0.39~~ 0.37 at $\Delta x \ge 10$ km to ~~0.65~~0.62–~~0.76 at $\Delta x \le 2.5$~~0.75 at $\Delta x \le 5$ km~~. At sub-kilometre spacing, part of this increase is an increase in cloud cover;~~ , and at 312 m ~~the cloud-cover~~ 0.32 of its 0.71 is cloud cover (0.52 with the LWP-threshold cover, which leaves an in-cloud part of ~~$S_{20\to50}$ is 0.33 with the maximum-overlap definition (0.53 with the LWP threshold). On the high~~0.19). At $\Delta x \ge 10$ km the cloud cover is 1.00 by both definitions, so a cloud-cover response is structurally absent there. The cloud-cover response also appears with N_d ~~branches cloud cover contributes less than 0.03. The temporal-sampling standard deviation is at most 0.03 at~~ -independent autoconversion

Table 4. Susceptibility with cloud-top N_d for sedimentation factor 1, hours 10–12. "All", "ic" and "cc" denote the all-sky value and its in-cloud and cloud-cover (maximum overlap) parts. "Indep." denotes N_d-independent autoconversion. $d\ln\tau/d\ln N_\mathrm{d}$ is for the moderate branch. S_{55} is the local susceptibility at a cloud-top N_d of 55 cm^{-3} and $N_{0.1}$ the cloud-top N_d (cm^{-3}) above which $|S| < 0.1$. Perturbed-twin and tile-bootstrap standard deviations are about 0.01 on the clean and moderate branches and 0.003 on the polluted branch at sub-kilometre spacing, and up to 0.03 at 5 km.

	$S_{20\to50}$			$S_{100\to200}$		$S_{200\to1000}$	indep.				
Δx	all	ic	cc	all	ic	all	$S_{100\to200}$	$S_{200\to1000}$	$d\ln\tau/d\ln N_\mathrm{d}$	S_{55}	$N_{0.1}$
40 km	0.37	0.37	0.00	0.25	0.25	0.08	−0.03	−0.01	0.55	0.38	664
20 km	0.37	0.37	0.00	0.25	0.25	0.08	−0.03	−0.01	0.54	0.37	653
10 km	0.36	0.36	0.00	0.22	0.22	0.06	−0.03	−0.01	0.52	0.35	260
5 km	0.62	0.52	0.10	0.21	0.21	0.01	0.05	0.01	0.51	0.44	257
2.5 km	0.72	0.49	0.23	0.11	0.11	−0.03	−0.01	−0.01	0.42	0.47	259
1.25 km	0.63	0.44	0.19	0.07	0.06	−0.03	−0.02	−0.01	0.38	0.43	112
625 m	0.75	0.51	0.24	0.03	0.02	−0.04	−0.06	−0.03	0.35	0.29	111
312 m	0.71	0.39	0.32	−0.05	−0.05	−0.04	−0.07	−0.05	0.29	0.06	54

at $\Delta x \le 2.5$ km~~and up to 0.1~~, where cover increases with N_d while in-cloud LWP decreases, which points to droplet sedimentation or radiative effects on thin, broken cloud. Near the observed RF02 droplet number, the local susceptibility at a cloud-top N_d of 55 cm^{-3} is 0.35–0.47 from 40 km to 1.25 km, 0.29 at ~~$\Delta x \ge 5$~~625 ~~km, where LWP oscillates on time scales of about 15~~m and 0.06 at 312 ~~min.~~

~~At~~ m. A single fit of $\ln$LWP against $\ln N_\mathrm{d}$ over all six levels gives a positive slope at every spacing (0.22 at 40 km and 0.10 at 312 m~~, the negative $S_{100\to200}$ corresponds to an LWP difference of 2.6~~), so the branch-wise values depend on the choice of branch boundaries, and the cloud-top N_d above which $|S| < 0.1$ decreases from about 660 ~~g m^{-2} (3.4~~cm^{-3} at 20–40 ~~%) between targets~~ km to 260 cm^{-3} at 2.5–10 km, 110 cm^{-3} at 1.25 km–625 m and 54 cm^{-3} at 312 m. The location of the LWP maximum is poorly constrained, because LWP at targets 100 and 200 cm^{-3} ~~, and $S_{200\to1000}$ to a difference of 4.1~~differs by only 3 ~~g m^{-2} between 200 and 1000 cm^{-3}. These differences are smaller than the LWP decrease of each run in one hour (Sect. 3.3), and initial-condition spread in domain-mean LWP of stratocumulus simulations is typically of a few per cent. Without an ensemble, the sign of these small susceptibilities at~~ % at 312 m ~~is therefore not established. The dependence on s is more systematic. At $\Delta x \le 2.5$ km with suppression on , $S_{200\to1000}$ is -0.01 to -0.02 for $s = 0.5$~~ and ~~-0.06 to -0.09 for $s = 2$ (Fig. (prev.)c).~~ 625 m.

The radiative meaning of these susceptibilities differs from their sign. The optical-depth susceptibility $d\ln\tau/d\ln N_\mathrm{d}$ stays positive on every branch at every spacing. On the moderate branch ~~, $S_{100\to200}$ increases with s at 2.5 and 1.25~~it decreases from 0.55 at 40 km ~~(from 0.09 to 0.23 and from 0.06~~ to ~~0.20) and decreases with s~~ 0.29 at 312 m~~(from -0.02 to -0.08). With suppression off, S is within ±0.13 of zero at $\Delta x \ge 10$~~, and on the clean branch it is 0.64 at 40 km ~~on all branches. At $\Delta x \le 1.25$~~and 0.66–0.77 at finer spacing. In the offline albedo split, the LWP and cloud-cover terms amplify the Twomey term

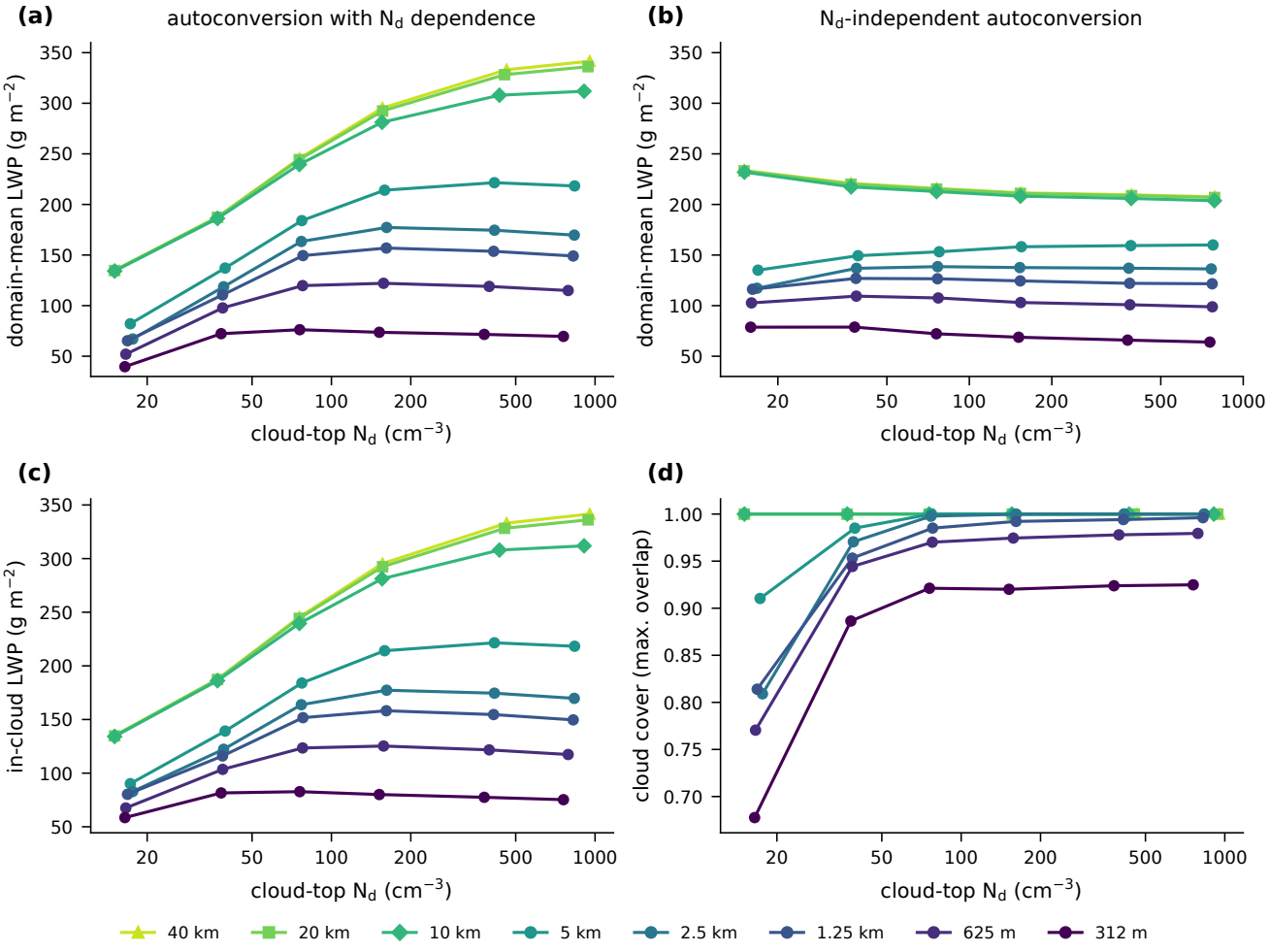


Figure 3. Response to cloud-top N_d at every grid spacing (sedimentation factor 1, hours 10–12). (a) Domain-mean LWP with N_d-dependent and (b) N_d-independent autoconversion. (c) In-cloud LWP and (d) maximum-overlap cloud cover with N_d-dependent autoconversion.

by 58 ~~km~~ % at 40 km on the moderate branch and offset it by 13 % at 312 m, while on the clean branch they amplify it by 91 % at 40 km and by a factor of 2.0–3.2 at $\Delta x \leq 5$ km, mostly through cloud cover. On the polluted branch the adjustment changes from $+16\,\%$ to between -5 and ~~$s \leq 1$ it is negative on the two high-N_d branches (-0.07 and -0.05 at~~ $-7\,\%$ of the Twomey term. The total potential albedo increase on the moderate branch is 0.079 at 40 km and 0.046 at 312 m ~~for $s = 1$).~~ A change in the sign of the LWP susceptibility at high N_d therefore reduces the implied brightening without reversing it, whereas the clean-branch increase at fine spacing amplifies it. Because clean regimes dominate the uncertainty in aerosol forcing (Carslaw et al., 2013) and the cloud response to N_d is nonlinear (Jia and Quaas, 2023), the clean-branch result is the more relevant one for forcing estimates.

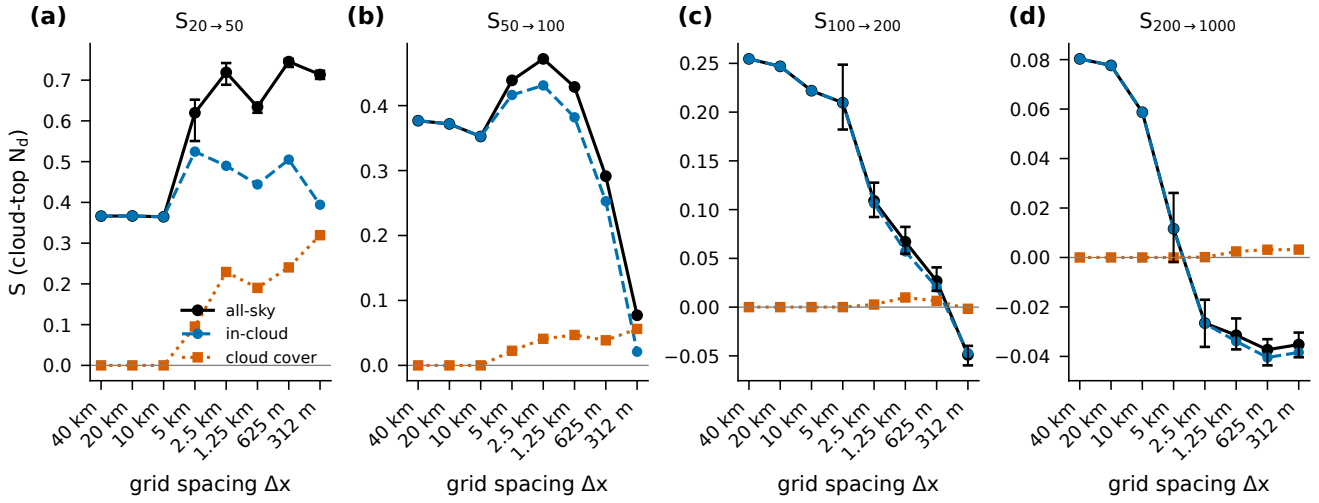


Figure 4. Susceptibility (cloud-top N_d) on the four branches against grid spacing, split into in-cloud and cloud-cover parts (sedimentation factor 1, N_d-dependent autoconversion, hours 10–12). Bars give the 5–95 % range of a 60 km tile bootstrap ($\Delta x \le 5$ km).

3.3 Temporal evolution

The ~~susceptibilities evolve throughout the 12 h~~ resolution dependence develops during the simulations (Fig. 5). ~~At $\Delta x \ge 10$ km, $S_{100\to200}$ increases from 0.12 in~~ In hours 2–4~~to 0.24~~, $S_{100\to200}$ is 0.10–0~~.27 in hours 10–12, as the high-N_d clouds continue to gain the water that low-N_d clouds lose to precipitation (Fig. 5d); at~~ .11 at every spacing from 40 to 2.5 km and ~~high N_d, LWP is still increasing by up to 15~~close to zero at $\Delta x \le 1.25$ g m^{-2} h^{-1} ~~in~~ km; by hours 10–12 ~~. At 1.25~~it has grown to 0.21–0.25 at 5–40 km and ~~625~~has become negative at 312 m, ~~$S_{100\to200}$ also increases slowly (from 0.00 to 0.07 and 0.03), whereas at 312~~while at 1.25 ~~m it decreases from 0.00 to −0.05~~km and 625 m it has increased slightly. The polluted branch ~~$S_{200\to1000}$ becomes more negative with time at all $\Delta x \le 2.5$~~is 0.04 at $\Delta x \ge 2.5$ km ~~, from about~~ and −0.01 at finer spacing in hours 2–4, and it diverges in the same way. At 40 km S oscillates with periods of about 1.5–2 h, so 2 h averages are needed. The cloud-top N_d of the LWP maximum falls below the highest N_d level after about 3 h at $\Delta x \le 1.25$ km~~and from 0.04 to between −0.03 and −0.04 in hours 10–12. The clean branch is already large in hours 2–4 at 625 and~~ and after about 9 h at 5 km, and at 312 ~~m (0.56–0.63) and at 1.25~~m it continues to decrease, from about 200 to 80 ~~km (0.51) and increases slowly. At $\Delta x \le 2.5$~~cm^{-3} between 5 and 12 h. The inverted-V shape is thus time-dependent, and the statement that the maximum lies near 100 cm^{-3} at 312 m refers to hour 12.

At $\Delta x \le 1.25$ km, LWP decreases in ~~98 % of the runs during hours 10–12,~~ nearly all runs throughout hours 4–12, and the coarse runs at high N_d are still thickening (by up to ~~4.4~~15 g m^{-2} h^{-1} ~~or 6at 40 % h^{-1}, and none of the susceptibilities at sub-kilometre spacing had levelled off at 12 h. The adjustment time scales in these runs are therefore longer than the~~

simulationkm), as in LES of non-precipitating stratocumulus, where LWP continues to adjust over a day or more (Glassmeier et al., 2021)
.

Figure (prev.) shows the decomposition of Eq. ((prev.)) at $s = 1$. At $\Delta x \geq 10$ km, the precipitation-suppression term at zero sedimentation is 0.34–0.39 on the moderate branch and 0.07–0.08 on the polluted branch, and the sedimentation term derived from the off runs is small (-0.02 to -0.01). The interaction term on the moderate branch is -0.07 to -0.08, because $S_{100\to200}$ with suppression on decreases strongly with s (from 0.35 to 0.20 between $s = 0.5$ and 2 transient simulations of precipitating shallow convection (Seifert et al., 2015). The phase plane of $d\mathrm{LWP}/dt$ against LWP at 40 km), whereas the off runs barely respond. This sensitivity to sedimentation at coarse spacing is larger than at 312 m ($dS_{\mathrm{on}}/ds = -0.10$ at 40 km and -0.04 at m (Fig. 5d) shows a common decline for all N_{d} that slows towards the end of the run. Exponential relaxation fits over hours 6–12 are well constrained only at 312 m). It is unlikely to act through entrainment, which changes by less than 0.6m, where they give time scales of 4–7 mmh and equilibrium LWP of about 45–55 s^{-1} between $s = 0.5$ and 2 at $\Delta x \geq 10$ $\mathrm{g\,m^{-2}}$ for $N_{\mathrm{d}} \geq 50\,\mathrm{cm^{-3}}$, 25–35 km (Fig. 8a). Deposition of cloud droplets at the surface is also unlikely to explain it, because the estimated deposition at the lowest model level exceeds 0.02% below the 12 mmh values. At coarser spacing the fits do not constrain an equilibrium within 24 $\mathrm{d^{-1}}$ only at $\Delta x \geq 10$h. The steady-state susceptibility at 312 km, target 20m implied by these fits, with 5–95 $\mathrm{cm^{-3}}$, $s = 2$ with suppression off (up to 0.25% intervals from a paired moving-block bootstrap, is -0.11 to -0.21 on the moderate branch in five of the six combinations of sedimentation and autoconversion setting (for $s = 1$ and N_{d}-dependent autoconversion -0.13, with an interval of -0.23 to $+0.06$), and -0.17 to $+0.06$ on the polluted branch. The negative moderate branch at 312 mmm is therefore more likely to strengthen than to vanish as the runs approach equilibrium, while the sign of the polluted branch at equilibrium is undetermined. Fast and slow time scales coexist, as in mixed-layer and LES analyses (Bretherton et al., 2010; Jones et al., 2014), and the inversion is far from its equilibrium. With a subsidence time scale of 74 $\mathrm{d^{-1}}$, or about 20h, w_{e}/D is about 1000 % of surface precipitation). The sensitivity to sedimentation at coarse spacing therefore depends on the autoconversion formulation, presumably through the vertical redistribution of cloud water relative to where rain forms, but we have not isolated its mechanism. In addition, because of the pivot dependence described in Sect. (prev.), the statement that precipitation suppression accounts for the positive response at coarse spacing holds for this design and should not be read as a pivot-independent attribution.

At finer spacing, the precipitation-suppression term on the moderate branch decreases to 0.12 at 5m at 40 km , 0.05–0.10 between 2.5and about 2200 km and 625 m and 0.04 at 312 m, and on the polluted branch it is 0.00–0.02 at $\Delta x \leq 5$compared with z_{i} of 860 and 1020 km. On the polluted branch at 2.5 and 1.25m at 12 km, the negative $S_{200\to1000}$ comes mainly from the interaction term (-0.04 and h. A susceptibility of about -0.05) and the residual (-0.01), while the sedimentation term derived from the off runs is near zero ($+0.01$ and -0.01). At 625after 11 m and 312h is weaker than the LWP adjustments found after a similar time in an LES ensemble of non-precipitating nocturnal stratocumulus (Chen et al., 2025).

3.4 Susceptibility and the base-state regime

Figure 6 plots S against the precipitation-to-entrainment-drying ratio of the lower-N_{d} run for all grid spacings, sedimentation factors and both autoconversion settings. With N_{d}-dependent autoconversion, $S_{100\to200}$ follows this ratio closely across the 24

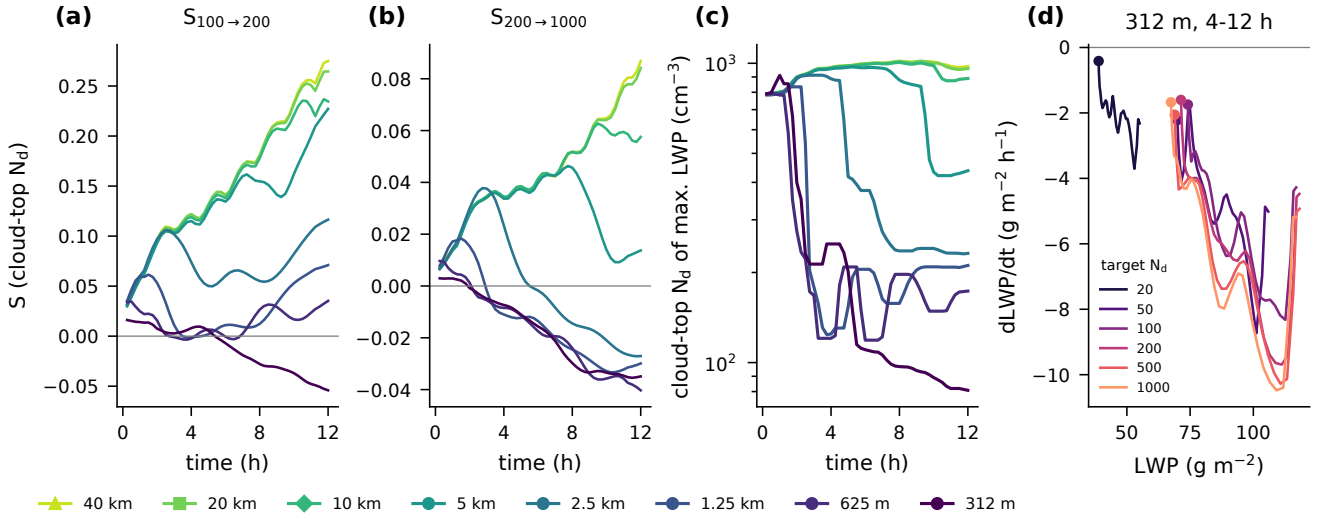


Figure 5. Temporal evolution (N_d-dependent autoconversion, sedimentation factor 1). (a) $S_{100\to200}$ and (b) $S_{200\to1000}$ from 15 min output (2 h running mean). (c) Cloud-top N_d of the LWP maximum (2 h running median); values near the top of the axis indicate a maximum at the highest N_d level. (d) Phase plane of $d\mathrm{LWP}/dt$ against LWP at 312 m for hours 4–12; dots mark hour 12.

combinations of grid spacing and sedimentation factor (correlation 0.92). The ratio alone explains 84 ~~m the sedimentation term (−0.016 and −0.024), the interaction (−0.015 at both) and the residual (−0.004 and −0.016)all contribute. On this branch, a larger sedimentation factor increases LWP more at target 200 than at 1000~~% of the variance of $S_{100\to200}$, and adding $\log\Delta x$ as a second predictor raises this only to 86 ~~cm^{-3}, so $S_{200\to1000}$ becomes more negative~~ %. This also explains why $S_{100\to200}$ increases with s ~~, consistent with the mechanism of Bretherton et al. (2007). At~~ at 2.5 and 1.25 km~~the moderate branch behaves differently, with positive sedimentation and interaction terms (both about 0.05), so that $S_{100\to200}$ increases with s.~~

~~On the clean branch at $\Delta x \le 5$~~, because faster sedimentation raises the base-state surface precipitation there and moves the runs along the same curve. On the middle branch the correlation is 0.70, and on the polluted branch 0.75, where grid spacing retains additional explanatory power (variance explained 57 ~~km, the sedimentation term is positive (0.13 at 5~~% by the ratio alone and 85 ~~km and 0.30–0.34 at $\Delta x \le 2.5$ km). At target $N_\mathrm{d} = 20\,\mathrm{cm}^{-3}$, cloud-top effective radii are 12–14 μm for $s = 1$ with suppression on, and a larger sedimentation factor lowers LWP most at the lowest N_d. At these droplet sizes, sedimentation mainly removes cloud water, as drizzle does, and its effect on entrainment is secondary. The clean-branch residual is large and negative at $\Delta x \le 2.5$ km (−0.16 to −0.28) , but it is the least well constrained term because S_off is convex in s (−0.11~~% with $\log\Delta x$), ~~0.00~~ consistent with the change in entrainment efficiency at constant, nearly zero precipitation. With N_d-independent autoconversion the relationship is weaker (correlations 0.45–0.65) and ~~0.34 for $s = 0.5$, 1 and 2 at 312 m). We interpret it only as an indication that pathways other than a linear sedimentation effect, possibly including radiative effective-radius effects on thin, broken clouds, contribute on this branch.~~ S is near zero over most of the range. On the clean branch the ratio has no explanatory power, because the response there is dominated by cloud cover at fine spacing.

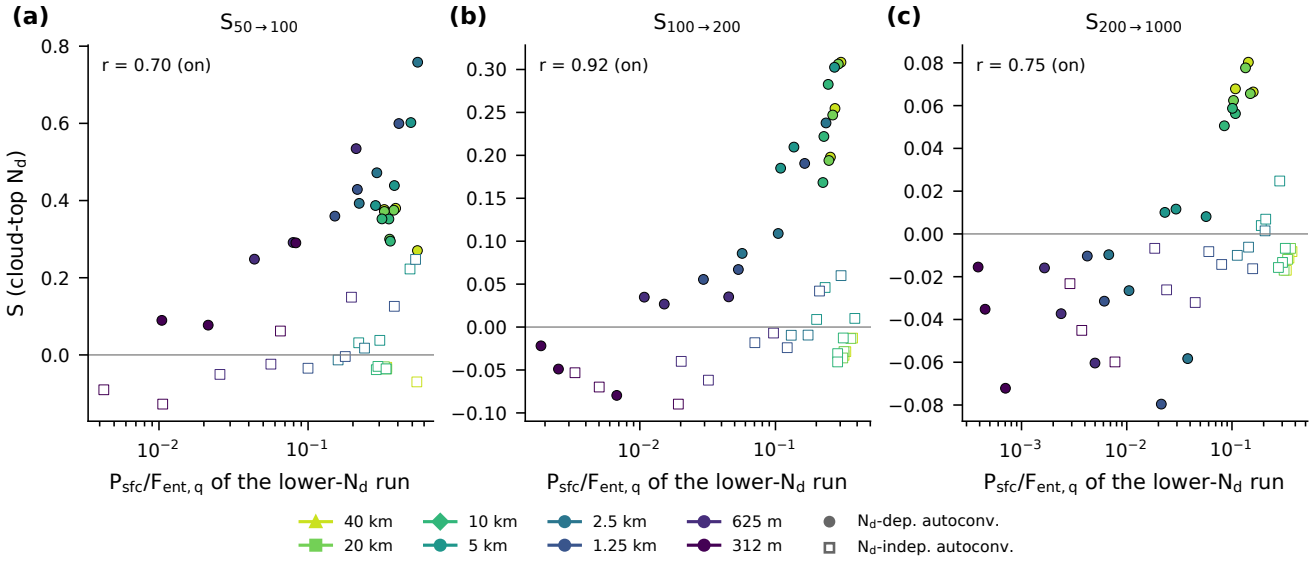


Figure 6. Susceptibility (cloud-top N_d, hours 10–12) against the ratio of surface precipitation to entrainment drying of the lower-N_d run, for all grid spacings and sedimentation factors, on the (a) middle, (b) moderate and (c) polluted branches. Filled circles: N_d-dependent autoconversion; open squares: N_d-independent. Correlations refer to N_d-dependent autoconversion.

~~The effect of precipitation suppression on LWP is limited by the size of the precipitation sink, and Fig. 7 shows that this sink decreases with grid spacing. Rain production (autoconversion plus accretion)~~ The base state changes with grid spacing because the coarse and fine clouds differ in depth, base height and turbulence. Rain production at target $200\,\mathrm{cm^{-3}}$ decreases from $0.55\,\mathrm{mm\,d^{-1}}$ at 40 km to $0.03\,\mathrm{mm\,d^{-1}}$ at 312 m ~~(Fig. 7a),~~ as the cloud thins from about 650 to ~~about~~ 300 m~~(Fig. 7d) and the KK2000 rates decrease steeply with cloud water. At target $20\,\mathrm{cm^{-3}}$ the decrease is smaller, from about 2.0 to $1.2\,\mathrm{mm\,d^{-1}}$. As cloud base rises, the fraction of rain production~~ , and the fraction that evaporates below cloud ~~increases, at target $200\,\mathrm{cm^{-3}}$~~ base increases from 0.40 ~~at 40 km~~ to 0.93 ~~at 312 m~~ (Fig. 7~~b), consistent with the importance of subcloud evaporation in drizzling stratocumulus (Wood, 2005; Wyant et al., 2007). At the same time the~~). Meanwhile the entrainment drying flux ~~$-\rho L_v w_e \Delta q_t$~~ roughly doubles, from 67 to $128\,\mathrm{W\,m^{-2}}$~~at target $200\,\mathrm{cm^{-3}}$ (Fig. 8c). The~~ , so the ratio of surface precipitation to entrainment drying ~~(Fig. 7c), which is the diagnostic that Ackerman et al. (2004) used to explain the sign of the LWP response, therefore decreases with grid spacing~~ decreases by a factor of about ~~4 at target $20\,\mathrm{cm^{-3}}$ (from 0.65 to 0.18) and by a factor of about~~ 300 at targets 200–$1000\,\mathrm{cm^{-3}}$ and of about 4 at target $20\,\mathrm{cm^{-3}}$. ~~The raining fraction of the domain (surface rain above 0.5~~Resolved variability partly compensates for the thinner cloud, since the autoconversion enhancement $\langle q_c^{2.47}\rangle / \langle q_c\rangle^{2.47}$ is 1.00 at $\Delta x \geq 10$ ~~mm~~km, 1.1–1.3 at $\Delta x \leq 5$ ~~$\mathrm{d^{-1}}$) is below 0.2 % at $\Delta x \leq 1.25$~~ km ~~for~~ and target $N_\mathrm{d} \geq 200\,\mathrm{cm^{-3}}$~~.~~ , and 1.4–1.9 at target $20\,\mathrm{cm^{-3}}$. The precipitation susceptibility S_R on the moderate branch increases from 0.9 at 40 km to 2.4–2.8 at $\Delta x \leq 2.5$ km, the coarse value reflecting the light, ubiquitous drizzle of the single-column limit (raining fraction 1.0 at target 20–$50\,\mathrm{cm^{-3}}$), which resembles the frequent light rain of GCMs (Michibata et al., 2016; Mülmenstädt et al., 2020). When sur-

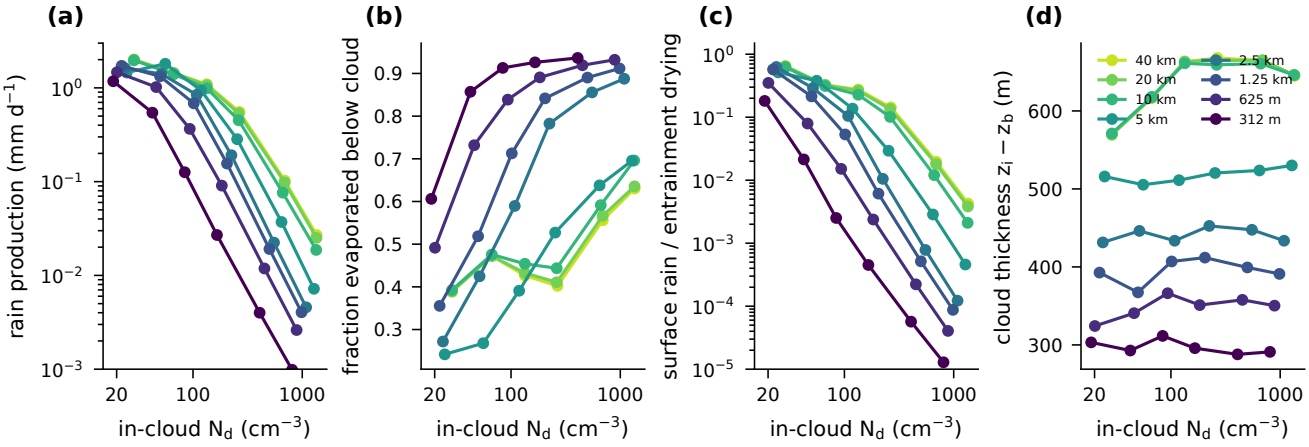


Figure 7. Precipitation against in-cloud N_d at every grid spacing (N_d-dependent autoconversion, sedimentation factor 1). (a) Column rain production. (b) Fraction of the production that evaporates before reaching the surface. (c) Ratio of surface precipitation to entrainment drying. (d) Cloud thickness $z_i - z_b$.

face precipitation is negligible, precipitation suppression has little effect on LWP, and the decrease of LWP with N_d associated with entrainment dominates (Ackerman et al., 2004; Wood, 2007; Hoffmann et al., 2020). In these simulations, the N_d at which this transition occurs depends on grid spacing ; in EAMxx, grid spacing determines which of these regimes the simulated cloud occupies.

3.5 Drizzle, sedimentation and entrainment

At target 200 cm^{-3} with suppression on Switching off the N_d dependence of autoconversion changes LWP through two parts of opposite sign, a rain sink and a drizzle effect on entrainment. In the mixed-layer budget at target $N_d \geq 200\,\mathrm{cm}^{-3}$, removing the heavy drizzle of the N_d-independent runs (that is, the entrainment rate is3.7the difference between the two autoconversion settings) contributes +4 to +6 mm% sh^{-1} at 40through the precipitation term at $\Delta x \geq 5$ km, 7.7 mm s^{-1} at 5 km, 6.8but only +0.2 to +1 mm% sh^{-1} at 2.5625 and 312 km, 7.4m. At the same time it raises w_e by 0.2 mm s^{-1} at 62520–40 m and 8.1km and by up to 1.4 mm s^{-1} at 312$\Delta x \leq 2.5$ m. The boundary-layer radiative cooling is 80–88 W m^{-2} in all these runs, so the change in w_e mostly reflects a change in entrainment efficiency. We define the ratio of entrainment warming to radiative cooling as

$$\eta = \frac{\rho c_p w_e \Delta\theta_l}{F_{\mathrm{rad}}}.$$

At $\Delta x \geq 10$ km , η is 0.45–0.67 km (Fig. 8b). At $\Delta x \leq 5$ km it exceeds one for target , consistent with drizzle stabilising the subcloud layer and reducing entrainment (Stevens et al., 1998; Ackerman et al., 2009). The difference between the two autoconversion settings therefore measures the precipitation-suppression effect net of a drizzle–entrainment effect, and its small value at fine spacing partly reflects cancellation. The factorial decomposition of Appendix A shows in addition that

the effect of sedimentation on S at coarse spacing depends on the autoconversion formulation, and that the partition between precipitation and sedimentation terms depends on the matching N_d ~~of 100–200~~of the N_d-independent configuration.

The offline cloud-top sedimentation flux at $s = 1$ is 0.4–1.5 ~~cm^{-3} and above, with the largest values at 5~~mm ~~km (1.07~~d^{-1}, with mass-weighted fall speeds of 1.0–1~~.25 for $N_\mathrm{d} \geq 100\,\mathrm{cm}^{-3}$) and at 312~~.2 m (1.2–1.3). When η exceeds one, entrainment warming exceeds radiative cooling, and with the surface sensible heat flux the mixed layer warms by 0.07–0.10~~cm K h~~s^{-1} ~~at 312~~at target 1000 ~~m. Entrainment drying at 312~~cm^{-3}, 1.8–2.3 ~~m (105–131~~cm ~~W~~s^{-1} at 200 ~~m^{-2} for $N_\mathrm{d} \geq 50\,\mathrm{cm}^{-3}$) likewise exceeds the prescribed latent heat flux of 93~~cm^{-3} and 3.8–5.3 ~~W~~cm ~~m^{-2}, whereas at $\Delta x \geq 10$~~s^{-1} at 20 ~~km it is 55–78~~cm^{-3}. Increasing s from 0.5 to 2 raises this flux by 0.5–1.5 ~~W~~mm ~~m^{-2} (Fig. 8c). The response of w_e to the sedimentation factor at fixed N_d~~ d^{-1} at target $N_\mathrm{d} \geq 200\,\mathrm{cm}^{-3}$ at all grid spacings (Fig. 8a)~~is largest at $\Delta x \leq 5$~~. At $\Delta x \leq 2.5$ km ~~and low N_d, where doubling the fall speed relative to $s = 0.5$ reduces~~ this increase is accompanied by a decrease of w_e by ~~1.1–2.4~~0.2–1.2 mm s^{-1} ~~at target $N_\mathrm{d} \leq 50\,\mathrm{cm}^{-3}$; at target $N_\mathrm{d} \geq 500\,\mathrm{cm}^{-3}$ the reduction is 0.2–0.8~~(regression slope -0.8 mm s^{-1} ~~. At~~ per mm d^{-1}, correlation -0.31), whereas at $\Delta x \geq 10$ km ~~the reduction is less than 0.6~~w_e does not respond (slope -0.03). The sedimentation–entrainmen coupling of Bretherton et al. (2007) thus operates at $\Delta x \leq 2.5$ km and is absent at coarse spacing, where the inversion is pinned. At low N_d the larger decrease of w_e with s (1.1–2.4 mm s^{-1} at ~~all N_d. This is the expected sign~~ $\Delta x \leq 5$ km) occurs together with large fall speeds and drizzle, so it is not a clean measure of the sedimentation–entrainment ~~feedback (Bretherton et al., 2007; Ackerman et al., 2009), and its larger magnitude~~ coupling. Because P3 sedimentation acts after SHOC's adjustment within each step and the cloud-top fields are written after P3, the time step (100 s at coarse and 15 s at fine spacing~~indicates that the coupling between sedimentation and entrainment is stronger where part of the boundary-layer circulation is resolved~~) may also affect how much cloud-top liquid is removed per step; cloud-top N_d relative to the target decreases with s by 8–12 % at 40 km but not at 312 m.

~~The mixed-layer budget (Sect. 2.5) expresses these fluxes as contributions to the LWP tendency~~

3.6 Decoupling and the mixed-layer budget

The degree of coupling depends on the metric (Fig. ~~10). In every run, radiative cooling and surface fluxes thicken the cloud, and entrainment and subsidence thin it. At 40~~9). The coarse runs have the largest moisture difference across the boundary layer ($\Delta q_t = 1.1$–1.5 ~~km, where the budget reproduces both the tendencies and their differences between N_d levels (Table (prev.)),~~ g kg^{-1}), produced by SHOC's down-gradient transport of the surface moisture flux, but their θ_l is well mixed, the ~~entrainment contribution is about −9~~buoyancy flux below cloud base stays at 40–110 ~~%~~% of its surface value and cloud base lies within 35 ~~h^{-1} at all N_d, and the precipitation contribution decreases in magnitude from −10 % h^{-1} at target 20 cm^{-3} to nearly zero at $N_\mathrm{d} \geq 500\,\mathrm{cm}^{-3}$. The increase of LWP with N_d at 40 km is therefore associated with the removal of the precipitation sink, consistent with Sect. (prev.).~~ m of the mixed-layer condensation level, so they are coupled by the Jones test and by the flux criterion. At 312 m ~~, the entrainment contribution is about three times larger (−27 to −32~~and target $N_\mathrm{d} \geq 100\,\mathrm{cm}^{-3}$, Δq_t is only 0.2–0.3 ~~%~~g ~~h~~kg^{-1} ~~for $N_\mathrm{d} \geq 50\,\mathrm{cm}^{-3}$), and it is partly offset by larger radiative (21–24~~, but the buoyancy-flux minimum below cloud base is 8–10 % of the surface value, cloud base lies 50–70 ~~h^{-1}) and surface-flux (12–13~~m above the mixed-layer condensation level, and the variance is concentrated below cloud. Refinement therefore increases decoupling by

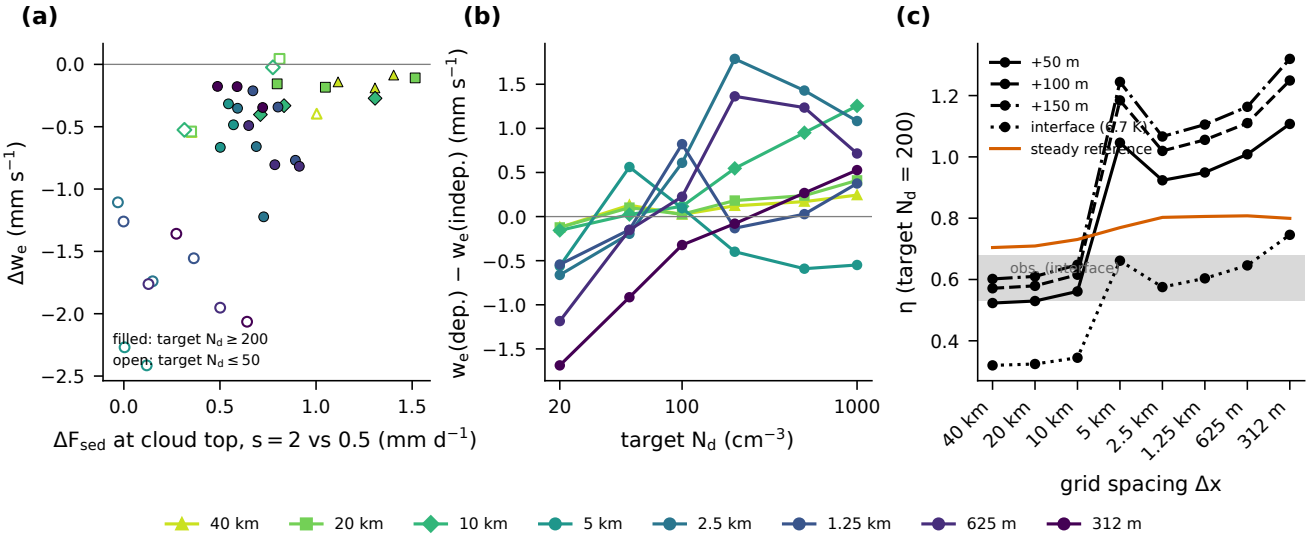


Figure 8. (a) Change in entrainment rate against change in the offline cloud-top sedimentation flux between sedimentation factors 2 and 0.5 (N_d-dependent autoconversion); filled symbols target $N_d \geq 200\,\mathrm{cm}^{-3}$, open symbols $\leq 50\,\mathrm{cm}^{-3}$. (b) Difference in entrainment rate between N_d-dependent and N_d-independent autoconversion (sedimentation factor 1). (c) η at target $N_d = 200\,\mathrm{cm}^{-3}$ for jumps taken 50, 100 and 150 m above z_i and for the specified interface jump, with the steady-state reference and the observed range for the interface definition.

the buoyancy-flux and cloud-base metrics while decreasing it by Δq_t. The buoyancy flux is not significantly negative below cloud base in any run, so the low-N_d states resemble cumulus rising into stratocumulus, without a stable subcloud layer.

By the combined criterion of Sect. 2.5, 97 % of the 2 h $^{-1}$) ~~contributions; the precipitation contribution is below 1~~windows at $\Delta x \geq 10$ ~~%~~km are coupled, 58 ~~h^{-1} in magnitude for $N_d \geq 100\,\mathrm{cm}^{-3}$. The budget sum is negative at all N_d (−2 to −5~~ % at 5 ~~h^{-1}), consistent in sign and magnitude with the simulated thinning (−4 to −6~~km, 11 ~~%~~% at 2.5 ~~h^{-1}). At 625~~km and at most 3 ~~m the budget sum at high N_d (−5.4 to −5.7~~ % at $\Delta x \leq 1.25$ ~~h^{-1}) is more negative than the simulated trend (−1.9 to −2.9 % h^{-1}). The budget therefore suggests that the general thinning at fine grid spacingis associated with~~ km. The mixed-layer budget (Fig. 10) is therefore used for attribution only at coarse spacing. There it reproduces the in-cloud LWP tendency in coupled windows with correlations of 0.72–0.76, and the increase of ~~entrainment relative to the radiative and surface forcing. At $\Delta x \leq 2.5$~~LWP with N_d at 40 km ~~, however, the net N_d dependence of the budget sum (about 2~~follows from the removal of the precipitation sink (−10 % h^{-1} at ~~312~~target 20 ~~m) is no larger than the budget error~~cm^{-3} and nearly zero at $\geq 500\,\mathrm{cm}^{-3}$) at an entrainment contribution that is constant in N_d. At 2.5 km and finer the correlation for in-cloud LWP falls to 0.47 and 0.07–0.32, and the entrainment ~~term itself varies non-monotonically with N_d (−29, −29, −27, −31 and −32~~contribution splits into a deepening part of +10 to +20 % h^{-1} ~~for targets 50 to 1000~~and a cloud-base part of −30 to −50 ~~cm^{-3}~~ % h^{-1} at 312 m~~)~~, so the net thinning is the small difference of large terms in a decoupled layer. The budget is consistent with the view that

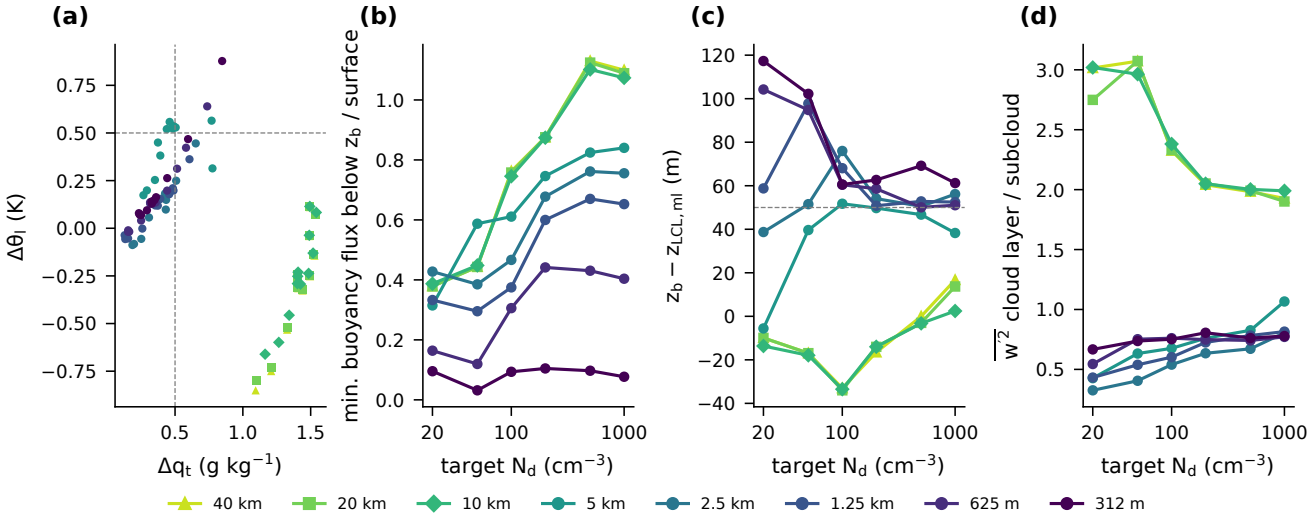


Figure 9. Decoupling metrics (sedimentation factor 1, hours 10–12). (a) $\Delta\theta_l$ against Δq_t between the highest and lowest quarters of the layer below the inversion base for both autoconversion settings, with the thresholds of Jones et al. (2011). (b) Minimum total buoyancy flux below cloud base relative to the surface value, (c) cloud base above the mixed-layer lifting condensation level, and (d) ratio of cloud-layer to subcloud vertical-velocity variance, for N_d-dependent autoconversion.

stronger entrainment raises cloud base faster than it deepens the layer at fine spacing, but it does not establish which process sets the small negative susceptibilities ~~at sub-kilometre spacing~~there.

3.7 Robustness

~~At $\Delta x \geq 10$ km, all vertical-velocity variance is parameterised (Fig. (prev.)a), and SHOC's boundary-layer variance lies between 0.28 and 0.41 m^2 s^{-2}. At 5 and 2.5 km this variance is 0.07~~The perturbed twins give standard deviations of the differences in S of 0.007–0~~.11 m^2 s^{-2}, and it decreases to 0.03 m^2 s^{-2} at 312 m, while the resolved variance increases. The resolved share of the total is 3–7 % at 5 km, 4–9 % at 2.5 km, 9–19 % at 1.25 km, 28–38 % at~~ .010 on the clean, middle and moderate branches and 0.002–0.004 on the polluted branch at 625 ~~m and 54–67 % at 312 m, so that a third of the variance remains subgrid even at 312 m. Part of this repartition is imposed by the closure. SHOC's mixing length is capped at the nominal grid spacing,~~ and ~~at~~ 312 ~~m the domain-mean mixing length is within 5 % of this cap at 90 % of the boundary-layer levels, compared with 25 % at 625 m and none at $\Delta x \geq 1.25$ km (Appendix B, Fig. B1a). At 312~~ m, ~~SHOC's eddy diffusivity is therefore directly limited by the grid spacing throughout the boundary layer. The large change between 10~~ so a single run has a noise of about 0.007 and 0.002, respectively. The tile bootstrap gives similar values at sub-kilometre spacing (0.006 and 0.003) and up to 0.03 at 5 ~~km, where the resolved share is still small, is also not explained by resolved turbulence. At 5 km SHOC's boundary-layer mean mixing length is about two thirds of its value at 10 km (Fig. B1a), even though the cap is not active,~~

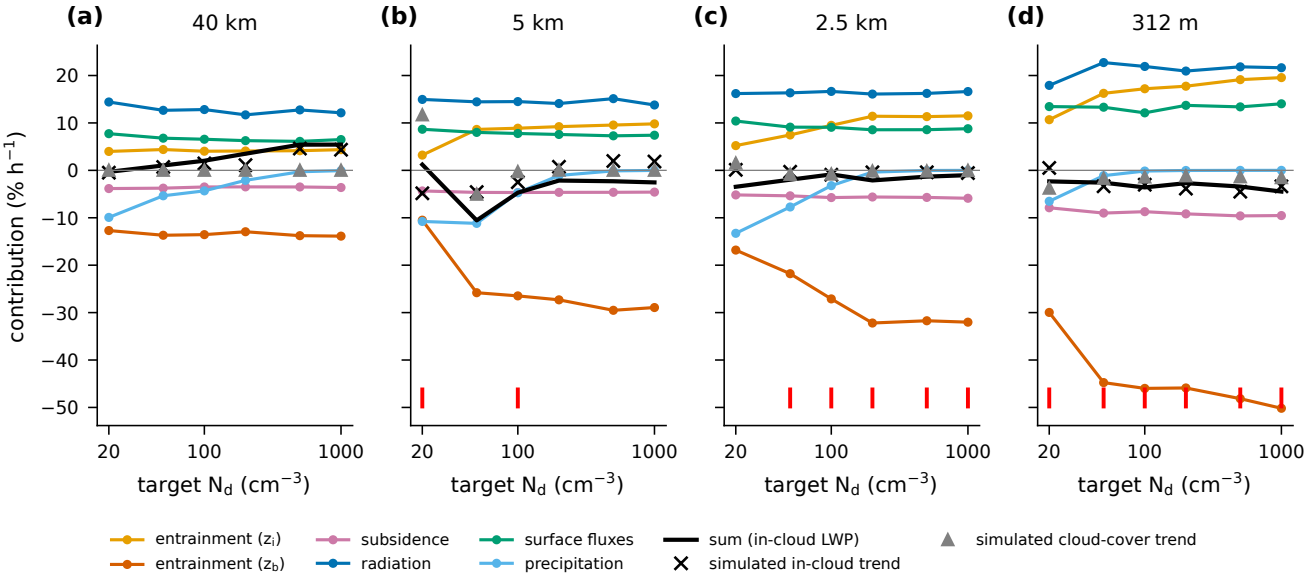


Figure 10. Mixed-layer budget of in-cloud LWP for hours 10–12 at (a) 40 km, (b) 5 km, (c) 2.5 km and (d) 312 m (N_d-dependent autoconversion, sedimentation factor 1), with the entrainment term split into deepening and cloud-base parts. Crosses and triangles are the simulated tendencies of in-cloud LWP and cloud cover. Red ticks mark windows that fail the coupling criterion, for which the attribution is not valid.

~~which points to a change in the SHOC solution itself, possibly coupled to the stripe-like circulations noted in Sect. (prev.). We have not diagnosed this transition further.~~

~~The resolved vertical-velocity skewness (Fig. (prev.)b), which is meaningful only at $\Delta x \leq 5$ km, is positive at low N_d (up to 1.9 at~~ km. The moderate-branch value at 312 ~~m for target 20 cm^{-3}) , indicating narrow, strong updrafts in a drizzle-decoupled boundary layer. At high N_d it is -0.7 to -1.0 at 2.5 and 1.25 km and -0.3 to -0.7 at 625 m, the sign expected for radiatively driven downdrafts in a well-mixed stratocumulus layer (Wood, 2012); the transition occurs at $N_d \approx 50$–100 cm^{-3}. At 312~~ m ~~the skewness stays positive at all N_d (0.09–0.24 for target $N_d \geq 100$ cm^{-3}) . With a third of the variance still subgrid and the mixing length at its cap, this may reflect near-grid-scale updrafts. Drizzle decouples the boundary layer at low N_d (Fig. (prev.)c). At $\Delta x \leq 5$ km, Δq_t is 1.1–1.6 g kg^{-1} at target 20 cm^{-3}~~ (-0.05, negative for all three sedimentation factors and both autoconversion settings) and ~~0.21–0.64~~the polluted-branch values at $\Delta x \leq 2.5$ ~~g kg^{-1} at 1000 cm^{-3}, consistent with below-cloud evaporation stabilising the subcloud layer (Stevens et al., 1998; Bretherton and Wyant, 1997; Jones et al., 2011). In the single-column limit, q_t decreases upward by 1.2–1.7 g kg^{-1} at all N_d across a deep cloud layer.~~

~~The LWP spectral peak changes less between~~ km are therefore well outside the internal noise. The moderate branch at 625 ~~and 312~~ m ~~than LWP itself (Fig. (prev.)d; Appendix D) . At 5 and 2.5 km the peak wavelengths of 48 and 30 km correspond to 6.4 and 8 physics-column spacings and are probably limited by the effective resolution of the dynamical core. At 625 and 312 m the peak lies at 13–22 km, an aspect ratio λ/z_i of 13–20. Observed closed mesoscale cells have aspect ratios of up~~

~~to about 30–40 for boundary layers about 1 km deep (Wood and Hartmann, 2006), and LES with large domains also produce~~
~~cells that are wider than these (Kazil et al., 2017; Zhou et al., 2018; Zhou and Bretherton, 2019).The sub-kilometre cells are~~
~~therefore on the narrow side of the observed range, and~~ (0.03) changes sign with the autoconversion setting, and the ~~2.5 km~~
~~value (33) agrees better with the observations, although it is probably resolution-limited.~~

~~Switching on 3D turbulence~~ location of the LWP maximum is not resolvable. The main threats to the fine-grid results are thus transience (Sect. 3.3) and structural choices, in particular the fixed vertical grid and the mixing-length cap, rather than internal variability. The 3D-turbulence option changes LWP by less than 1.5 ~~$\mathrm{g\,m^{-2}}$ ($< 1.5\,\%$), cloud cover by less than 0.007 %~~ and w_e by ~~-0.13 to $+0.02$~~less than 0.13 $\mathrm{mm\,s^{-1}}$ ~~at 625 m and 312 m~~ (Appendix C). ~~The 3D turbulence option therefore has a negligible effect on the results, which is expected because SHOC's horizontal diffusion is small compared with the horizontal dissipation of the dynamical core.~~

4 Discussion

4.1 One mixed-layer argument

~~None of the bulk quantities converges with grid spacing (Fig. (prev.)). At fixed diagnosed N_d, LWP decreases by 10–23 % from 1.25 km to 625 m and by a further 21–40 % from 625 to 312 m, the larger decreases occurring at higher N_d. The maximum-overlap cloud cover and~~ The results can be summarised by a single chain. Grid spacing changes the entrainment rate (Sect. 3.1); the entrainment rate sets the depth and base height of the cloud and hence its rain production and the fraction of rain that evaporates (Sect. 3.4); and the resulting ratio of precipitation to entrainment drying sets the moderate-branch susceptibility, as in the ~~cloud thickness also continue to decrease. For LWP at target $N_\mathrm{d} \geq 50\,\mathrm{cm^{-3}}$, the change over the last halving of the grid spacing exceeds that over the previous one by a factor of 1.3–2.0, so the sequence is not in an asymptotic range and an extrapolation to zero grid spacing is not meaningful. The entrainment rate is non-monotonic, with a minimum at 2.5–1.25 km, as expected in the grey zone where parameterised and resolved contributions trade places (Honnert et al., 2020; Boutle et al., 2014) . The clean-branch susceptibility (0.65–0.76~~ mixed-layer arguments of Ackerman et al. (2004), Wood (2007) and Hoffmann et al. (2020) . At coarse spacing the sink removed by precipitation suppression is large and the response is positive; at ~~$\Delta x \leq 2.5$~~fine spacing there is little rain to suppress, the sedimentation–entrainment coupling becomes active and the response is near zero or slightly negative. Most of the resolution dependence of $S_{100\to200}$ is thus a dependence on the simulated cloud regime, which grid spacing controls in EAMxx; at fixed regime, grid spacing adds little explanatory power on this branch. The near-zero susceptibility at fine spacing reflects an absent precipitation sink, which differs from the compensation between processes that Stevens and Feingold (2009) described as buffering. Two parts of the response do not follow this chain, namely the cloud-cover response on the clean branch, which is absent at coarse spacing by construction and which dominates the radiative relevance at fine spacing, and the polluted branch, which retains a dependence on grid spacing at nearly zero precipitation.

The first link of the chain is the least physical. Entrainment changes abruptly between 10 and 5 ~~km) varies less with grid spacing than the bulk quantities, but it varies with s from 0.56 to 0.97 at 312 m. The vertical grid (about 50 m at the inversion,~~ compared with about km as the 5 ~~m in the RF02 LES; Ackerman et al., 2009) was held fixed, and vertical resolution at the~~

inversion is known to control entrainment in LES and in SCREAM (Stevens et al., 2005a; Bogenschutz et al., 2023b). The sign of $S_{100\Rightarrow200}$ km runs leave a state pinned at the inversion, and at 3125 m may change with further horizontal or vertical refinement.

Satellite estimates of LWP adjustments use N_d retrieved from cloud-top effective radius and optical depth under adiabatic assumptions (Grosvenor et al., 2018). The model-diagnosed cloud-top N_d used here is not such a retrieval. It is 44–72km the resolved flow is dominated by near-grid-scale motions. At $\Delta x \leq 2.5$ % of the in-cloud layer value at $\Delta x \geq 10$ km and 77–95 % at 312 m, and because this ratio varies little with N_d at a given grid spacing , susceptibilities computed with cloud-top N_d differ from those computed with layer N_d by at most 0.10, 0.04 and 0.01 on the three branches (Fig. E1a). This agreement follows largely from the near-constant ratio and says nothing about retrieval errors.

Within each domain CCN are horizontally uniform, so the spatial covariation of LWP and cloud-top N_d arises from cloud processes. We regressed column $\ln$LWP on column cloud-top $\ln N_d$ over cloudy columns at each 15 min instant in hours 10–12, restricting the analysis to $\Delta x \leq 5$ km because the columns at coarser spacingare nearly identical. The slopes range from -7.9 to $+3.4$, with standard errors over instants of at most 0.8 and correlations between -0.86 and $+0.56$ (Fig. E1c). At 2.5–5 km and low N_d they are negative, because collision–coalescence depletes N_d in the thickest, drizzling columns (Fig. E1b). At 625 and 312 m and target $N_d \geq 50\,\mathrm{cm}^{-3}$ they are positive (0.4–3.3), because about half of the inferred entrainment flux is not carried by resolved or parameterised turbulence, the inversion spans two to three levels and cloud-top N_d is reduced in thin, evaporating cloud edges. At 312height varies by less than one level. LES studies show that entrainment converges from above as the inversion is resolved (Stevens and Bretherton, 1999; Stevens et al., 2005a; Pressel et al., 2017; Mellado et al., . Our sub-kilometre runs, with fixed 50 m and target $200\,\mathrm{cm}^{-3}$, for example, the within-scene slope is $+1.9$, whereas $S_{100\Rightarrow200} = -0.05$. That cloud-controlled N_d variability produces spatial LWP–N_d relationships unrelated to the causal response is well established (Gryspeerdt et al., 2019; Arola et al., 2022; Mülmenstädt et al., 2024b; Mahfouz et al., 2024). The additional result here is that the sign of this covariation changes with grid spacing, because the relative importance of drizzle depletion in thick cloud and evaporation in thin cloud edges depends on how much of the cloud field is resolved.

At $\Delta x \geq 10$ km, where the simulations approximate single-column physics, entrainment in EAMxx is nearly insensitive to droplet sedimentation, the clouds are thick enough to precipitate substantially, and LWP increases monotonically with N_d. This behaviour corresponds to the precipitation-suppression mechanism of Albrecht (1989). At finer grid spacing, entrainment increases ($\eta > 1$), and the boundary layer warms, dries and deepens. The cloud thins and its base rises, so that rain production decreases and more of the rain evaporates below cloud. In addition, w_e becomes more sensitive to droplet sedimentation (Fig. 8a). Together these changes shift the LWP maximum from $N_d \geq 1000$ to about $100\,\mathrm{cm}^{-3}$. This picture is consistent with mixed-layer and LES arguments in which the turning point of the inverted-V relationship is set by the competition between precipitation and entrainment (Ackerman et al., 2004; Wood, 2007; Hoffmann et al., 2020, 2024). In EAMxx, the balance between these processes depends on horizontal grid spacing across the grey zone.

The extent to which this dependence reflects grey-zone physics rather than specific features of the configuration is not yet clear. Three model-specific factors co-vary with grid spacing in our experiment. SHOC's mixing length is capped at the grid spacing, and this cap is active throughout the boundary layer m levels, a capped mixing length and an entrainment efficiency of

2.0–2.6 at 312 m. The physics time step and radiation interval decrease with grid spacing. The largest change in most quantities occurs between 10 and , are therefore unlikely to lie on a path towards the LES solution, and a converged answer probably requires entrainment to decrease again with vertical refinement. The grey zone for stratocumulus in EAMxx with SHOC begins at about 5 km, where the resolved variance is still small and the circulations appear partly numerical. Any of these could contribute to the trends attributed above to grid spacing, and Sect. 4.3 lists the experiments that would separate them. and remedies would include scale-aware partitioning of turbulence (Boutle et al., 2014; Efstathiou and Beare, 2015; Honnert et al., 2011) and vertical resolution at the inversion consistent with the horizontal grid (Bogenschutz et al., 2023b).

The weak negative branch at 312

4.2 Relation to other models and to LES

The coarse configuration reproduces features of GCM behaviour, namely no cloud-cover response, light ubiquitous drizzle and a positive response driven by autoconversion, although with a 100 m deserves particular caution. Its magnitude ($S \approx -0.04$ to -0.05) is an order of magnitude smaller than the steady-state adjustments inferred from LES of non-precipitating stratocumulus (Glassmeier et al., 2021) and smaller than the observed slopes, which are themselves upper bounds (Gryspeerdt et al., 2019) . Possible reasons include the transience of the runs, the absence of a droplet-size dependence of cloud-top evaporation in SHOC's saturation adjustment (Wang et al., 2003; Hoffmann and Feingold, 2019), the s time step and 50 m vertical spacing at the inversion, and the humidity of the overlying air, which controls the strength of entrainment drying (Ackerman et al., 2004; Chen et al., 20 ; in our runs the q_t jump at the inversion is 5.8–6.5 g kg^{-1}.

The closest prior study is that of Terai et al. (2020) , who compared an ultra-parameterised multiscale model with levels it is not a GCM column. The comparison with Mülmenstädt et al. (2024a), who found an active sedimentation–entrainment coupling in E3SM single-column physics, indicates that whether this coupling appears in single-column physics depends on the turbulence scheme and its vertical representation of the inversion, which is pinned at a model level in our coarse runs. Terai et al. (2020) found weaker LWP increases at 250 m cloud-resolving grids against a conventional multiscale model with than at 4 km grids. They found a weaker LWP increase with aerosol at the finer resolution, caused by weaker LWP in a multiscale model because of weaker increases in non-raining clouds and a smaller raining fraction. Our results show , the same tendency for a single case in a doubly periodic configuration, in that the raining fraction and the precipitation-suppression term decrease with grid spacing, and LWP increases with N_d only weakly or not at all in the non-raining runsat sub-kilometre spacingas in our runs. Sato et al. (2018) attributed the LWP decrease in their LWP decreases in a global cloud-system-resolving model to explicitly simulated condensation and evaporationresponses to aerosol. That pathway is , a pathway absent from EAMxx, in which condensation is diagnosed by SHOC's saturation adjustment, so the mechanisms in the two models differ.

The comparison with Mülmenstädt et al. (2024a) is instructive. Using E3SM single-column physics, they found that switching sedimentation off removed the LWP decrease and that doubling it made the response more negative, and they concluded that GCM-type physics can represent the sedimentation–entrainment coupling. In our coarse runs, which use SHOC rather than the turbulence scheme of . The 312 m polluted-branch value after 11 h (-0.04) is of a similar magnitude to the weak nocturnal LWP decreases at high N_d in LES by Hill et al. (2009) and to the weak slopes expected for weakly and non-precipitating

stratocumulus by Hoffmann et al. (2025), and weaker than the LES ensemble of Chen et al. (2025) at a similar time after perturbation. Possible reasons include the absence of a size-dependent evaporation–entrainment effect in SHOC (Wang et al., 2003; Hoffma , the ~~E3SM GCM, a~~ 50 ~~m vertical grid and a 100 s time step, entrainment is nearly insensitive to sedimentation, while the susceptibility with precipitation suppression on is sensitive to it through an interaction with autoconversion. The two results are not in conflict, because the configurations differ, but they show that the sedimentation–entrainment coupling in single-column physics depends on the turbulence scheme and its numerical settings. Our coarse runs are therefore not representative of GCM columns, which use much longer time steps, coarser vertical grids and different turbulence and microphysics schemes, and we do not draw conclusions about GCM behaviour from them. Likewise, GSRMs with other turbulence closures, such as Smagorinsky-type or scale-aware schemes that blend boundary-layer and LES closures (Boutle et al., 2014; Efstathiou and Beare, 2015) , may behave differently in the grey zone. Within EAMxx, the susceptibility at 2.5–5 km (0.11–0.20 on the moderate branch, about 40–75 % of~~ m levels, drizzle at the lower N_d levels, and the ~~40~~moderately humid free troposphere of RF02 (Δq_t of 5.8–6.5 ~~km value)lies between the values at 40~~g ~~km and 312 m. These results suggest that estimates of aerosol–cloud adjustments from GSRMs should be accompanied by resolution-sensitivity tests, and they motivate grey-zone-aware turbulence closures. The factorial approach also offers process-level targets, the precipitation-suppression term and the sensitivity to sedimentation, that could help to reduce equifinality in constraining aerosol–cloud parameterisations (Stanford et al., 2026)~~ kg^{-1} at the inversion). Because solar heating buffers the nocturnal decline of LWP (Zhang et al., 2024; Chen et al., 2024; Sandu et al., 2008 and the meteorological controls on albedo susceptibility differ between regions and times of day (Zhang and Zuidema, 2019; Zhang and Fei , a diurnally averaged susceptibility cannot be inferred from these runs, and the nocturnal values are likely to overstate the thinning.

~~Using model-diagnosed cloud-top N_d instead of layer-mean N_d changes S by at most 0.1 in these simulations , but this does not test retrieval biases, and satellite analyses also use in-cloud LWP at pixel scale rather than domain means. The~~ Satellite-style analyses of these simulations are hypothetical because the case is nocturnal; they are summarised in Appendix E. With horizontally uniform CCN, within-scene ~~covariation of LWP and~~ variability accounts for less than 1 % of the variance of cloud-top $\ln N_d$, so pooled composites reproduce the between-run slope, whereas a retrieval-like adiabatic N_d ~~with uniform aerosol spans a range of slopes much wider than the susceptibilities and changes sign with grid spacing. This adds a model-based illustration to the concerns raised by Ghan et al. (2016), Gryspeerdt et al. (2019), Arola et al. (2022) and Mahfouz et al. (2024) that spatial co-variability is not a measure of the adjustment, and it cautions against evaluating the LWP adjustments of kilometre-scale models with satellite statistics that do not account for cloud-controlled N_d variability~~is about twice the model value and half of its variance is within-scene noise correlated with LWP, which lowers the pooled slope substantially. Selecting overcast, non-raining columns alone shifts the slope by up to 0.1. These results are consistent with Gryspeerdt et al. (2019), Quaas et al. (2020), Arola et al. (2022), Gryspeerdt et al. (2022b) and Kokkola et al. (2025), and with large-domain LES analyses of satellite-style relationships (Costa-Surós et al., 2020; Dipu et al., 2022), and they do not add a separate conclusion.

4.3 Limitations and future experiments

The study simulates ~~a single~~ one nocturnal, drizzling case with constant forcing~~and no shortwave radiation. The diurnal cycle modulates adjustments (Sandu et al., 2008), and the humidity above the inversion controls the sign of the entrainment response (Ackerman et al., 2004; Chun et al., 2023).~~ , and hours 10–12 have no counterpart in nature. Each configuration was run once ~~. The quoted spread in S reflects temporal sampling only, and the small negative susceptibilities at sub-kilometre spacing (-0.02 to -0.09) require confirmation with initial-condition ensembles of three to five members, for example at target N_d of 100, 200 and 1000 cm^{-3} at 2.5 km and 312 m~~with a common initial perturbation; the perturbed twins bound the internal noise but not the structural uncertainty. After 12 h the sub-kilometre runs are still ~~thinning by 4–6 % h^{-1}, and their susceptibilities have not levelled off, so a subset should be extended to 24 h or longer (Glassmeier et al., 2021; Bretherton et al., 2010).~~

~~The surface fluxes were prescribed. A bulk estimate that scales the latent heat flux with $q_\mathrm{s}(T_\mathrm{s}) - q_t$ and the sensible heat flux with $T_\mathrm{s} - T_\mathrm{air}$, calibrated to the prescribed fluxes and the initial mixed layer with a sea surface temperature of 292 K, suggests that interactive fluxes would have been 88–103 and 6–10 W m^{-2} at 312 m instead of 93 and 16 W m^{-2}, and 77–79 and 17–20 W m^{-2} at $\Delta x \geq 10$ km. The feedback would thus reduce the warming of the sub-kilometre boundary layers and would slightly increase their moisture supply, but at 312 m the estimated latent heat flux remains below the entrainment drying of 105–131 W m^{-2}. The estimate assumes fixed transfer coefficients and wind speed and should be tested with interactive surface fluxes over a fixed sea surface temperature. The absence of Coriolis force and wind nudging allowed the initial shear to decay, which may affect shear-driven entrainment in a resolution-dependent way.~~

~~The finest grid (312 m, with physics columns 469 m apart) is much coarser than the RF02 LES, and the vertical spacing at the inversion (about 50 m) was held fixed. Neither LWP nor $S_{100\to200}$ converged, so the results at 312 m describe the trend of a grey-zone model and not a converged answer. A converged reference with the same microphysics, for example an LES with KK2000 autoconversion at the same N_d levels, is not available. The physics time step and radiation interval co-varied with grid spacing, and~~ adjusting, and the steady-state values depend on exponential extrapolation. The vertical grid was fixed, SHOC's mixing length was capped at the grid spacing~~at 312 m. Experiments with a common time step at 40, 5~~ , the time step co-varied with grid spacing, the radiation scheme lacks the above-inversion balancing term of the protocol, and ~~2.5 km, with a relaxed mixing-length cap at 625 and 312 m, and with refined vertical spacing near the inversion would separate these effects from grid spacing. The decomposition of the susceptibility depends on the matching N_d of the suppression-off configuration and on an extrapolation to zero sedimentation. Simulations with $s = 0$, with a second matching N_d (for example 200 or 1000 cm^{-3}), or with a design in which the N_d seen by autoconversion and the N_d seen by sedimentation and radiation are varied independently, would give pivot-independent partial derivatives. The 240 km domain contains 11–19 cells at sub-kilometre spacing, and at 5 and 2.5 km the cell scale is probably limited by the effective resolution of the dynamical core.~~

the absence of Coriolis force and wind nudging allowed the shear to decay. CCN were prescribed and ~~horizontally~~ uniform, without aerosol processing ~~, scavenging or replenishment, which can matter for drizzle-driven transitions (Kazil et al., 2011; Berner et al., 2~~ ~~. Condensation and evaporation are computed by SHOC's~~ (Kazil et al., 2011; Berner et al., 2013; Goren et al., 2019; Yamaguchi et al., 201 , condensation was diagnosed by saturation adjustment, ~~so the droplet-size dependence of cloud-top evaporation that contributes~~

to the entrainment pathway in LES (Wang et al., 2003; Bretherton et al., 2007; Hoffmann and Feingold, 2019) is absent. Autoconversion follows and autoconversion followed KK2000 without a subgrid variance enhancement, so at coarse spacing it uses the mean in-cloud cloud water, whereas at fine spacing resolved variability partly provides such an enhancement; other autoconversion schemes would change the precipitation-suppression term (Morrison et al., 2020). Surface precipitation includes the flux of cloud droplets reaching the surface, which we estimate to be negligible except in subgrid enhancement. The decomposition into precipitation and sedimentation terms depends on the matching N_d of the coarse runs at target 20 N_d-independent configuration and on an extrapolation in s. The most informative future experiments would be a paired-perturbation design in which the runs are spun up at a common N_d and then perturbed and integrated for 24–36 cm^{-3} with $s = 2$ and suppression off. Finally, because the P3 sedimentation diagnostics were not usable, the sedimentation flux at cloud top was not diagnosed, and the role of sedimentation was inferred from the sedimentation-factor experimentsh with several seeds; a 312 m pair with 10–15 m levels near the inversion; a 36 h set with a diurnal cycle; simulations with a sedimentation factor of zero and with a second matching N_d for the N_d-independent configuration; tests with a common time step and a relaxed mixing-length cap; and an LES reference with the same microphysics.

5 Conclusions

We analysed 288 doubly periodic EAMxx simulations of the nocturnal, drizzling DYCOMS-II RF02 drizzling stratocumulus with horizontal stratocumulus under constant forcing, with grid spacings from 40 km to 312 m, target six N_d from 20 to 1000 cm^{-3}, three cloud-droplet sedimentation factors , and the N_d dependence of autoconversion switched on or offlevels, three sedimentation factors and two autoconversion settings. Our conclusions are as follows.

1. The LWP response to N_d depends strongly on grid spacing. At 10–40After 10–12 km LWP increased monotonically with N_d ($S_{100\to200} = 0.24$–0.27). With decreasing grid spacing the LWP maximum moved to lower N_d, reaching about 100h, the moderate-branch LWP susceptibility decreased from 0.25 at 40 cm^{-3} km to -0.05 at 312 m , where $S_{100\to200} = -0.05$ and $S_{200\to1000} = -0.04$ in single, still-evolving realisations.

2. At coarse spacing, removing the N_d dependence of autoconversion removed the positive response, but the sensitivity to sedimentation interacted strongly with autoconversion, and the partition between the two depends on the matching N_d of the experiment design. With decreasing grid spacing, and the precipitation-suppression contribution on the moderate branch decreased from 0.34polluted branch became weakly negative at $\Delta x \le 2.5$ km (23 of 24 configurations), whereas the clean-branch susceptibility increased from 0.37 to 0.62–0.39 to 0.04, and the gain in LWP from precipitation suppression at a diagnosed .75, largely through cloud cover. The optical-depth susceptibility stayed positive, so the implied brightening is reduced at high N_d and amplified at low N_d of 500 cm^{-3} decreased from 110 to 6 $g\,m^{-2}$at fine spacing.

3. ~~At sub-kilometre spacing the weak decrease of LWP at high N_d was also present without the N_d dependence of autoconversion and became steeper with faster sedimentation on the polluted branch, consistent with a sedimentation–entrainment feedback. Its sign is not established, given its small magnitude, the transience of the runs and the absence of ensembles.~~

4. ~~The ratio of entrainment warming to radiative cooling increased from about 0.5 at coarse spacing to 1.2–1.3 at~~ The moderate-branch susceptibility tracks the ratio of surface precipitation to entrainment drying of the base state across grid spacings and sedimentation factors. In EAMxx, grid spacing acts mainly by setting the cloud regime through the entrainment rate, which determines cloud depth, cloud base, rain production and subcloud evaporation.

5. The resolution dependence develops over the run and the fine-grid runs are still adjusting. The 312 m ~~, entrainment drying exceeded the prescribed surface moisture supply, and the clouds thinned, lifted and produced less rain, most of which evaporated below cloud. A mixed-layer budget associates the general thinning with this increase in entrainment , but it has little skill for the N_d dependence at $\Delta x \leq 2.5$ km.~~ moderate-branch value is well outside the internal noise and more likely to strengthen than to vanish at equilibrium, whereas the equilibrium sign of the polluted branch is undetermined.

6. ~~Neither LWP nor its N_d response converged, and LWP decreased by 21–40~~Part of the increase of entrainment with refinement is implicit mixing at an inversion spanning two to three 50 ~~% between 625 and 312~~m levels, the 5 ~~m at fixed diagnosed N_d. SHOC's mixing length was capped by the grid spacing at~~ km runs are dominated by near-grid-scale circulations, and the 312 m ~~, the time step co-varied with grid spacing, and 3D turbulence had a negligible effect. The 625 m runs were closest to the RF02 observations at hours 4–6, but no grid spacing matched all observed quantities.~~

7. ~~Susceptibilities computed with model-diagnosed cloud-top N_d agreed closely with those computed with layer N_d, whereas within-scene LWP–N_d slopes with uniform aerosol ranged from -7.9 to $+3.4$ and changed sign with grid spacing. In EAMxx, the simulated LWP adjustment in drizzling stratocumulus thus depends on how the boundary-layer turbulence is partitioned between the grid and the turbulence scheme, as well as on the microphysics. Initial-condition ensembles,~~ runs do not approximate LES. Extending these results to physical estimates requires vertical refinement at the inversion, longer and diurnally forced simulations, ~~interactive surface fluxes, tests of time step and mixing-length limits, refined vertical grids, a second matching N_d for the suppression-off configuration and additional cases are needed to determine how far these results generalise.~~ and paired-perturbation ensembles.

Appendix A: Factorial decomposition

~~Table (prev.) lists the skill of the mixed-layer budget for the LWP tendency at~~ For each grid spacing ~~, for the raw tendencies over all 2 h windows from 2 to 12 h and for the anomalies relative to the mean over the six~~ and branch we fitted $S_{\mathrm{dep}}(s) = a_{\mathrm{dep}} + b_{\mathrm{dep}} s$ and $S_{\mathrm{ind}}(s) = a_{\mathrm{ind}} + b_{\mathrm{ind}} s$ for N_d ~~levels at fixed grid spacing, sedimentation factor, suppression setting and window.Figure (prev.) shows the anomalies. At 40, 20~~ -dependent and ~~5 km the budget reproduces both the tendencies and their~~ N_d ~~dependence. At $\Delta x \leq 2.5$~~ -independent autoconversion over $s = 0.5$, 1 and 2, and decomposed $S_{\mathrm{dep}}(1) = (a_{\mathrm{dep}} - a_{\mathrm{ind}}) + b_{\mathrm{ind}} + (b_{\mathrm{dep}} - b_{\mathrm{ind}}) + a_{\mathrm{ind}}$

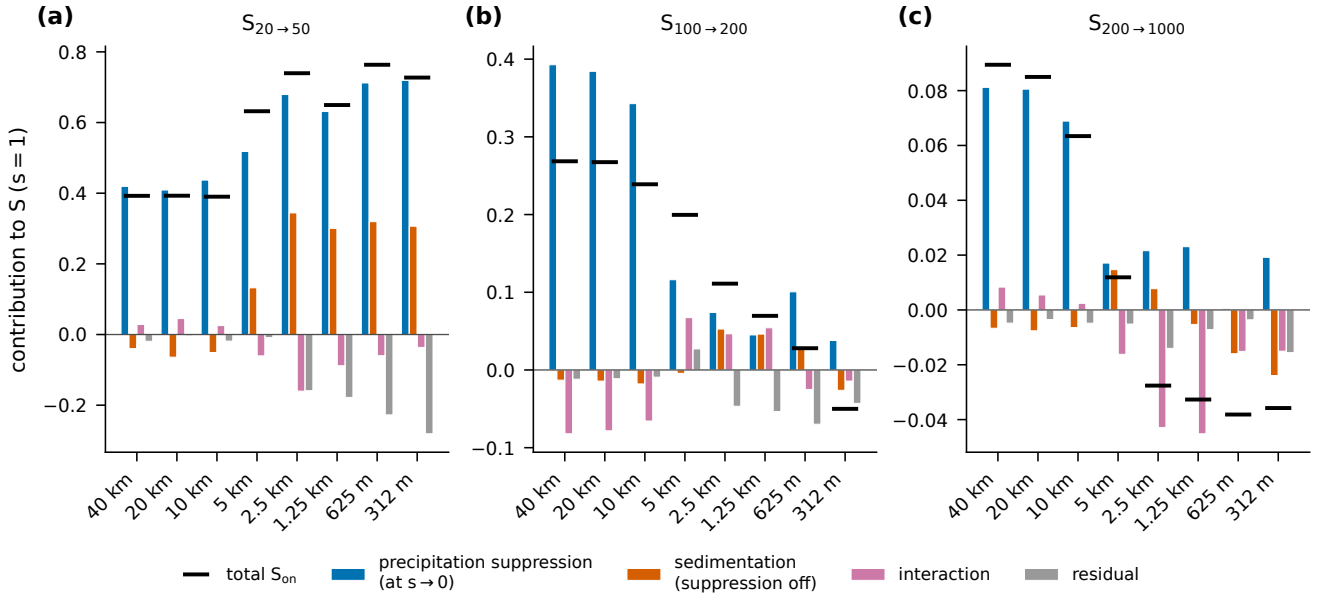


Figure A1. Factorial decomposition of the susceptibility with N_d-dependent autoconversion at sedimentation factor 1 (black bars) on the (a) clean, (b) moderate and (c) polluted branches.

into an autoconversion term at zero sedimentation, a sedimentation term, an interaction and a residual (Fig. A1, computed with the layer N_d of the first version). At $\Delta x \geq 10$ km the ~~correlation for the anomalies in hours 10–12 is between −0.02 and 0.57, and the root-mean-square error is comparable to or larger than the standard deviation of the anomalies. The largest departures in the raw tendencies occur where the boundary layer is decoupled and cloud base departs from the mixed-layer lifting condensation level. Using the simulated dz_b/dt instead of the estimate from the lifting condensation level degrades the agreement, because z_b is quantised by the vertical grid~~autoconversion term on the moderate branch is 0.34–0.39 and the interaction -0.07 to -0.08, because $S_{100\to200}$ decreases strongly with s only with N_d-dependent autoconversion; at finer spacing the autoconversion term decreases to 0.04–0.12. Because the N_d-independent runs drizzle heavily at high N_d, the extrapolation to $s = 0$ has one residual degree of freedom and S is convex in s on the clean branch, the partition depends on the design and should not be read as a pivot-independent attribution. Section 3.4 offers a simpler interpretation, in which sedimentation moves the base state along the regime curve.

Appendix B: SHOC mixing length and activation floor

Figure B1a shows the domain-mean SHOC mixing length ~~in hours 10–12~~ at target 200 cm^{-3}. At $\Delta x \geq 10$ km it reaches about 1600 m ~~in the upper boundary layer~~near 0.85 z_i, about twice the boundary-layer depth, because the grid-length cap is not active. At 5 km to 1.25 km it peaks at 900–1100 m ~~near 0.85 z_i and is about~~ and is 600–800 m below. At 625 m it is 480–610 m, close

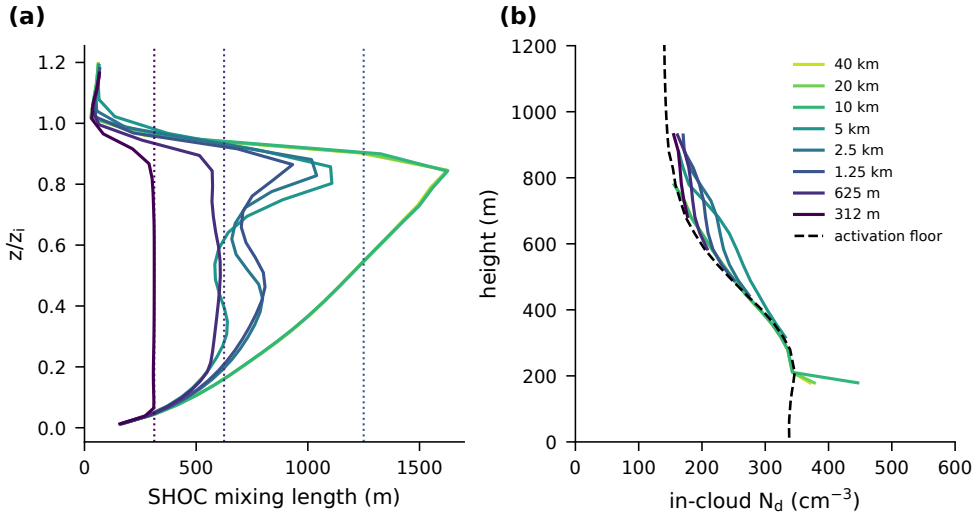


Figure B1. (a) Domain-mean SHOC mixing length against z/z_i and (b) in-cloud N_d with the activation floor (dashed), for target $N_\mathrm{d} = 200\,\mathrm{cm}^{-3}$, hours 10–12, N_d-dependent autoconversion and sedimentation factor 1. Dotted lines in (a) mark the grid-spacing cap at 1.25 km, 625 m and 312 m.

to the cap~~of 625 m~~, and at 312 m it equals the cap ~~of 312 m~~ nearly everywhere in the boundary layer. Figure B1b shows that the in-cloud N_d ~~profiles and~~ follows the activation floor~~for target 200 cm^{-3}. The in-cloud N_d follows the floor~~, which decreases with height~~because the prescribed CCN number does, so the diagnosed in-cloud N_d depends on the height of the cloud~~ ; between 6 and 12 h the layer N_d decreases by about 3 % h^{-1} at $\Delta x \leq 625$ m as the cloud rises into air with less CCN.

Appendix C: 3D turbulence twins

Figure C1 shows the ~~difference between~~ differences between the 71 runs with 3D turbulence on and ~~off for the 36 pairs at 625 m and the 35 completed pairs at 312 m~~their partners. LWP changes by between -1.0 and $+1.4\,\mathrm{g\,m}^{-2}$, cloud cover by between -0.005 and $+0.007$, and the entrainment rate by between -0.13 and $+0.02\,\mathrm{mm\,s}^{-1}$~~. The only systematic signal is a slight reduction of entrainment. These differences are an order of magnitude smaller than those between successive grid spacings or between N_d levels.~~ ; the root-mean-square relative LWP difference in hours 10–12 is 0.6 %.

Appendix D: LWP spectra and distributions

~~Figure D1 shows the LWP spectra and column-LWP distributions referred to in Sect. (prev.). The spectral peak shifts from~~ The LWP spectral peak (Fig. D1) lies at 48 km at 5 km ~~to~~ and about 30 km at 2.5 km~~and to~~ , 6.4 and 8 physics-column spacings, which suggests that it is limited by the effective resolution of the dynamical core, and at 13–22 km at 625 and 312 m~~. The~~

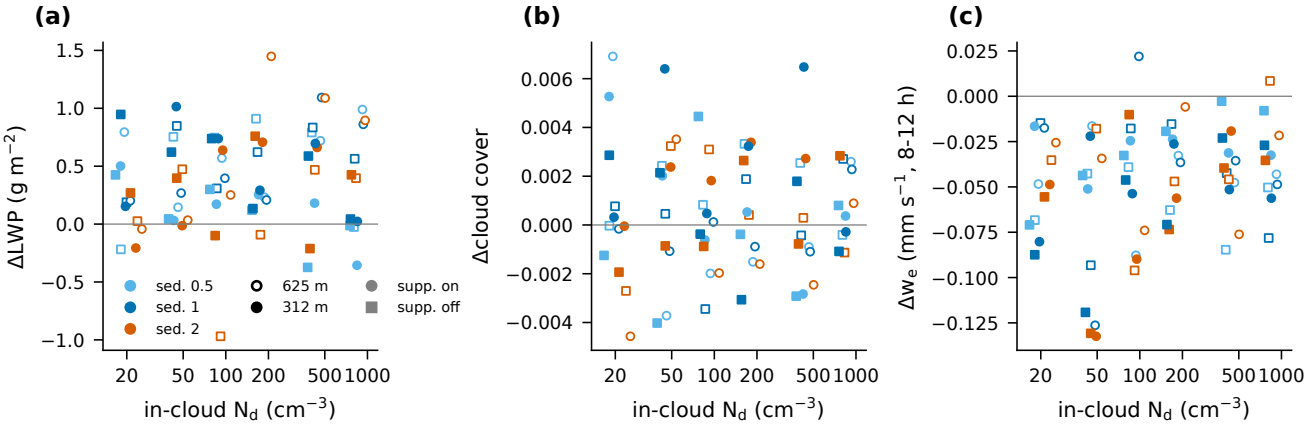


Figure C1. Difference (3D turbulence on minus off) in (a) LWP, (b) cloud cover (LWP above $10\,\mathrm{g\,m^{-2}}$) and (c) entrainment rate at 625 m (open symbols) and 312 m (filled symbols), coloured by sedimentation factor; circles: N_d-dependent, squares: N_d-independent autoconversion.

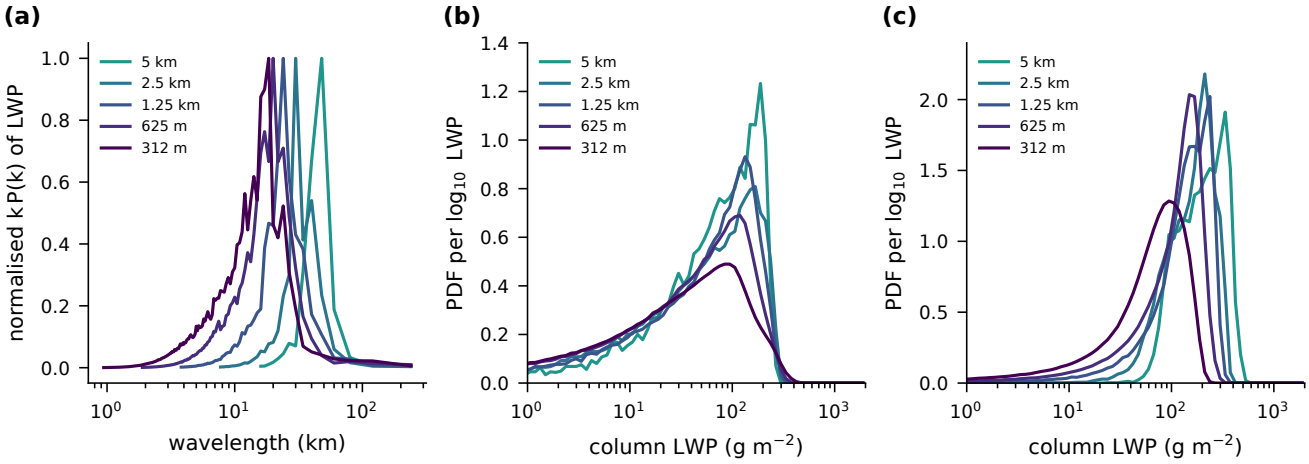


Figure D1. (a) Normalised radially averaged LWP spectra (hours 10–12) for target $N_\mathrm{d} = 200\,\mathrm{cm^{-3}}$. (b, c) Distributions of column LWP over hours 10–12 for target N_d of 20 and $1000\,\mathrm{cm^{-3}}$. N_d-dependent autoconversion, sedimentation factor 1.

~~LWP distributions broaden and shift to lower values as the grid is refined,~~ , an aspect ratio of 13–20 that is on the narrow side of observed closed cells (Wood and Hartmann, 2006; Glassmeier and Feingold, 2017). Pockets of open cells were observed during RF02 and in LES of the case (Stevens et al., 2005b; Savic-Jovcic and Stevens, 2008), so the broken low-N_d fields are plausible, whereas the reduced cloud cover at high N_d at 312 m has no observational counterpart in this case.

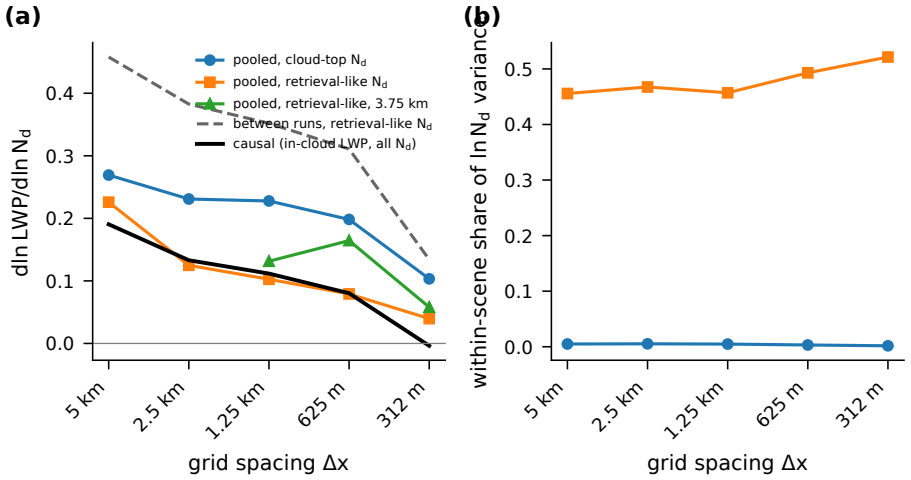


Figure E1. (a) Pooled-composite slope of $\ln \mathrm{LWP}$ on $\ln N_\mathrm{d}$ for cloud-top N_d and retrieval-like N_d (native and 3.75 km), the between-run slope with retrieval-like N_d, and the slope of in-cloud LWP for all columns (N_d-dependent autoconversion, sedimentation factor 1, hours 10–12). (b) Within-scene share of the pooled $\ln N_\mathrm{d}$ variance.

Appendix E: Satellite-style analysis

To illustrate how satellite-style estimators would perform if these clouds could be observed, we pooled the cloudy columns of the six N_d runs at each grid spacing with equal weight per run and computed the regression of column $\ln \mathrm{LWP}$ on column $\ln N_\mathrm{d}$ (Fig. E1). Columns were selected with a satellite-like filter: cloud-top liquid fraction above 0.9, LWP above $20\,\mathrm{g\,m^{-2}}$, cloud-top effective radius below $15\,\mu$m and optical depth above 3. The retrieval-like N_d follows the adiabatic formula of Grosvenor et al. (2018) applied to column LWP and cloud-top effective radius, at native resolution and after averaging to a common 3.75 km scale (McComiskey and Feingold, 2012). With model cloud-top N_d, the within-scene share of the $\ln N_\mathrm{d}$ variance is below 1 % (within-scene standard deviation of $\ln N_\mathrm{d}$ at most 0.12, correlations between -0.66 and $+0.57$), so the pooled slope (0.10 at 312 m to 0.27 at ~~target $N_\mathrm{d} = 20\,\mathrm{cm^{-3}}$ they develop a large population of clear or nearly clear columns~~ 5 km) equals the between-run slope; it exceeds the slope of in-cloud LWP for all columns (0.00 to 0.19) because the filter removes broken and drizzling columns. With retrieval-like N_d, which is about twice the model value, the within-scene share is 46–52 % and the pooled slope (0.04–0.23) is only 25–50 % of the between-run slope (0.13–0.46), illustrating the dilution and bias by retrieval errors correlated with LWP (Jia et al., 2019; Arola et al., 2022; Kokkola et al., 2025; Jia et al., 2022). Because activation in EAMxx does not depend on updraft speed, the model lacks the updraft-driven covariation of N_d and LWP found in nature, and the within-scene relationships should not be generalised to observations.

Code and data availability. The model code is the E3SM branch `mahf708/eamxx/p3-cloud-sed-factor` at https://github.com/mahf708/E3SM/tree/mahf708/eamxx/p3-cloud-sed-factor, which contains E3SM pull request 9017 (the cloud-top output operator) and the P3 sedimentation-factor option. Run scripts and all analysis scripts used to produce the tables and figures are in the `sed-precsupp-study` directory of the branch https://github.com/mahf708/scmlib/tree/sed-precsupp-study. Model output and the derived tables are available from the corresponding contact on request.

Author contributions. Naser Mahfouz and Johannes Mülmenstädt designed the experiments, and Naser Mahfouz implemented the model modifications and performed the simulations. The author, an AI system (Claude, Anthropic), designed and carried out the analysis and wrote the manuscript at the direction of Naser Mahfouz and Johannes Mülmenstädt. Human review of the manuscript is pending.

Competing interests. The author declares that there are no competing interests.

Acknowledgements. Naser Mahfouz and Johannes Mülmenstädt (Pacific Northwest National Laboratory) designed and directed this study, and Naser Mahfouz is the contact person for correspondence. The simulations used the Energy Exascale Earth System Model (E3SM) and resources of the National Energy Research Scientific Computing Center (NERSC), a U.S. Department of Energy Office of Science User Facility, on the Perlmutter system. The analysis and the text were produced by an AI system (Claude, Anthropic); all analysis scripts are provided so that the results can be verified.

References

Ackerman, A. S., Kirkpatrick, M. P., Stevens, D. E., and Toon, O. B.: The impact of humidity above stratiform clouds on indirect aerosol climate forcing, Nature, 432, 1014–1017, https://doi.org/10.1038/nature03174, 2004.

Ackerman, A. S., vanZanten, M. C., Stevens, B., Savic-Jovcic, V., Bretherton, C. S., Chlond, A., Golaz, J.-C., Jiang, H., Khairoutdinov, M., Krueger, S. K., Lewellen, D. C., Lock, A., Moeng, C.-H., Nakamura, K., Petters, M. D., Snider, J. R., Weinbrecht, S., and Zulauf, M.: Large-Eddy Simulations of a Drizzling, Stratocumulus-Topped Marine Boundary Layer, Monthly Weather Review, 137, 1083–1110, https://doi.org/10.1175/2008mwr2582.1, 2009.

Albrecht, B. A.: Aerosols, Cloud Microphysics, and Fractional Cloudiness, Science, 245, 1227–1230, https://doi.org/10.1126/science.245.4923.1227, 1989.

Arola, A., Lipponen, A., Kolmonen, P., Virtanen, T. H., Bellouin, N., Grosvenor, D. P., Gryspeerdt, E., Quaas, J., and Kokkola, H.: Aerosol effects on clouds are concealed by natural cloud heterogeneity and satellite retrieval errors, Nature Communications, 13, https://doi.org/10.1038/s41467-022-34948-5, 2022.

Beare, R. J.: A Length Scale Defining Partially-Resolved Boundary-Layer Turbulence Simulations, Boundary-Layer Meteorology, 151, 39–55, https://doi.org/10.1007/s10546-013-9881-3, 2014.

Bellouin, N., Quaas, J., Gryspeerdt, E., Kinne, S., Stier, P., Watson-Parris, D., Boucher, O., Carslaw, K. S., Christensen, M., Daniau, A.-L., Dufresne, J.-L., Feingold, G., Fiedler, S., Forster, P., Gettelman, A., Haywood, J. M., Lohmann, U., Malavelle, F., Mauritsen, T., McCoy, D. T., Myhre, G., Mülmenstädt, J., Neubauer, D., Possner, A., Rugenstein, M., Sato, Y., Schulz, M., Schwartz, S. E., Sourdeval, O., Storelvmo, T., Toll, V., Winker, D., and Stevens, B.: Bounding Global Aerosol Radiative Forcing of Climate Change, Reviews of Geophysics, 58, https://doi.org/10.1029/2019rg000660, 2020.

Berner, A. H., Bretherton, C. S., Wood, R., and Muhlbauer, A.: Marine boundary layer cloud regimes and POC formation in a CRM coupled to a bulk aerosol scheme, Atmospheric Chemistry and Physics, 13, 12 549–12 572, https://doi.org/10.5194/acp-13-12549-2013, 2013.

Bogenschutz, P. A. and Krueger, S. K.: A simplified PDF parameterization of subgrid-scale clouds and turbulence for cloud-resolving models, Journal of Advances in Modeling Earth Systems, 5, 195–211, https://doi.org/10.1002/jame.20018, 2013.

Bogenschutz, P. A., Eldred, C., and Caldwell, P. M.: Horizontal Resolution Sensitivity of the Simple Convection-Permitting E3SM Atmosphere Model in a Doubly-Periodic Configuration, Journal of Advances in Modeling Earth Systems, 15, https://doi.org/10.1029/2022ms003466, 2023a.

Bogenschutz, P. A., Lee, H.-H., Tang, Q., and Yamaguchi, T.: Combining regional mesh refinement with vertically enhanced physics to target marine stratocumulus biases as demonstrated in the Energy Exascale Earth System Model version 1, Geoscientific Model Development, 16, 335–352, https://doi.org/10.5194/gmd-16-335-2023, 2023b.

Bogenschutz, P. A., Clevenger, T. C., Bradley, A. M., Caldwell, P. M., Beydoun, H., Mahfouz, N., Keen, N. D., Guba, O., Bertagna, L., Foucar, J., Zhang, J., and Donahue, A. S.: High Performance, High Fidelity: A GPU-Accelerated Doubly-Periodic Configuration of the Simple Cloud-Resolving E3SM Atmosphere Model Version 1 (DP-SCREAMv1), Journal of Advances in Modeling Earth Systems, 17, e2025MS005 127, https://doi.org/10.1029/2025ms005127, 2025.

Bolton, D.: The Computation of Equivalent Potential Temperature, Monthly Weather Review, 108, 1046–1053, https://doi.org/10.1175/1520-0493(1980)108<1046:tcoept>2.0.co;2, 1980.

Boutle, I. A., Eyre, J. E. J., and Lock, A. P.: Seamless Stratocumulus Simulation across the Turbulent Gray Zone, Monthly Weather Review, 142, 1655–1668, https://doi.org/10.1175/mwr-d-13-00229.1, 2014.

Bretherton, C. S. and Wyant, M. C.: Moisture Transport, Lower-Tropospheric Stability, and Decoupling of Cloud-Topped Boundary Layers, Journal of the Atmospheric Sciences, 54, 148–167, https://doi.org/10.1175/1520-0469(1997)054<0148:mtltsa>2.0.co;2, 1997.

Bretherton, C. S., Macvean, M. K., Bechtold, P., Chlond, A., Cotton, W. R., Cuxart, J., Cuijpers, H., Mhairoutdinov, M., Kosovic, B., Lewellen, D., Moeng, C.-H., Siebesma, P., Stevens, B., Stevens, D. E., Sykes, I., and Wyant, M. C.: An intercomparison of radiatively driven entrainment and turbulence in a smoke cloud, as simulated by different numerical models, Quarterly Journal of the Royal Meteorological Society, 125, 391–423, https://doi.org/10.1002/qj.49712555402, 1999.

Bretherton, C. S., Blossey, P. N., and Uchida, J.: Cloud droplet sedimentation, entrainment efficiency, and subtropical stratocumulus albedo, Geophysical Research Letters, 34, https://doi.org/10.1029/2006gl027648, 2007.

Bretherton, C. S., Uchida, J., and Blossey, P. N.: Slow Manifolds and Multiple Equilibria in Stratocumulus-Capped Boundary Layers, Journal of Advances in Modeling Earth Systems, 2, https://doi.org/10.3894/james.2010.2.14, 2010.

Caldwell, P. M., Terai, C. R., Hillman, B., Keen, N. D., Bogenschutz, P., Lin, W., Beydoun, H., Taylor, M., Bertagna, L., Bradley, A. M., Clevenger, T. C., Donahue, A. S., Eldred, C., Foucar, J., Golaz, J.-C., Guba, O., Jacob, R., Johnson, J., Krishna, J., Liu, W., Pressel, K., Salinger, A. G., Singh, B., Steyer, A., Ullrich, P., Wu, D., Yuan, X., Shpund, J., Ma, H.-Y., and Zender, C. S.: Convection-Permitting Simulations With the E3SM Global Atmosphere Model, Journal of Advances in Modeling Earth Systems, 13, https://doi.org/10.1029/2021ms002544, 2021.

Carslaw, K. S., Lee, L. A., Reddington, C. L., Pringle, K. J., Rap, A., Forster, P. M., Mann, G. W., Spracklen, D. V., Woodhouse, M. T., Regayre, L. A., and Pierce, J. R.: Large contribution of natural aerosols to uncertainty in indirect forcing, Nature, 503, 67–71, https://doi.org/10.1038/nature12674, 2013.

Chen, Y.-C., Christensen, M. W., Stephens, G. L., and Seinfeld, J. H.: Satellite-based estimate of global aerosol–cloud radiative forcing by marine warm clouds, Nature Geoscience, 7, 643–646, https://doi.org/10.1038/ngeo2214, 2014.

Chen, Y.-S., Zhang, J., Hoffmann, F., Yamaguchi, T., Glassmeier, F., Zhou, X., and Feingold, G.: Diurnal evolution of non-precipitating marine stratocumuli in a large-eddy simulation ensemble, Atmospheric Chemistry and Physics, 24, 12 661–12 685, https://doi.org/10.5194/acp-24-12661-2024, 2024.

Chen, Y.-S., Prabhakaran, P., Hoffmann, F., Kazil, J., Yamaguchi, T., and Feingold, G.: Magnitude and timescale of liquid water path adjustments to cloud droplet number concentration perturbations for nocturnal non-precipitating marine stratocumulus, Atmospheric Chemistry and Physics, 25, 6141–6159, https://doi.org/10.5194/acp-25-6141-2025, 2025.

Chun, J.-Y., Wood, R., Blossey, P., and Doherty, S. J.: Microphysical, macrophysical, and radiative responses of subtropical marine clouds to aerosol injections, Atmospheric Chemistry and Physics, 23, 1345–1368, https://doi.org/10.5194/acp-23-1345-2023, 2023.

Costa-Surós, M., Sourdeval, O., Acquistapace, C., Baars, H., Carbajal Henken, C., Genz, C., Hesemann, J., Jimenez, C., König, M., Kretzschmar, J., Madenach, N., Meyer, C. I., Schrödner, R., Seifert, P., Senf, F., Brueck, M., Cioni, G., Engels, J. F., Fieg, K., Gorges, K., Heinze, R., Siligam, P. K., Burkhardt, U., Crewell, S., Hoose, C., Seifert, A., Tegen, I., and Quaas, J.: Detection and attribution of aerosol–cloud interactions in large-domain large-eddy simulations with the ICOsahedral Non-hydrostatic model, Atmospheric Chemistry and Physics, 20, 5657–5678, https://doi.org/10.5194/acp-20-5657-2020, 2020.

de Roode, S. R., Frederikse, T., Siebesma, A. P., Ackerman, A. S., Chylik, J., Field, P. R., Fricke, J., Gryschka, M., Hill, A., Honnert, R., Krueger, S. K., Lac, C., Lesage, A. T., and Tomassini, L.: Turbulent Transport in the Gray Zone: A Large Eddy Model Intercomparison Study of the CONSTRAIN Cold Air Outbreak Case, Journal of Advances in Modeling Earth Systems, 11, 597–623, https://doi.org/10.1029/2018ms001443, 2019.

Dipu, S., Schwarz, M., Ekman, A. M. L., Gryspeerdt, E., Goren, T., Sourdeval, O., Mülmenstädt, J., and Quaas, J.: Exploring Satellite-Derived Relationships between Cloud Droplet Number Concentration and Liquid Water Path Using a Large-Domain Large-Eddy Simulation, Tellus B: Chemical and Physical Meteorology, 74, 176, https://doi.org/10.16993/tellusb.27, 2022.

Donahue, A. S., Caldwell, P. M., Bertagna, L., Beydoun, H., Bogenschutz, P. A., Bradley, A. M., Clevenger, T. C., Foucar, J., Golaz, C., Guba, O., Hannah, W., Hillman, B. R., Johnson, J. N., Keen, N., Lin, W., Singh, B., Sreepathi, S., Taylor, M. A., Tian, J., Terai, C. R., Ullrich, P. A., Yuan, X., and Zhang, Y.: To Exascale and Beyond—The Simple Cloud-Resolving E3SM Atmosphere Model (SCREAM), a Performance Portable Global Atmosphere Model for Cloud-Resolving Scales, Journal of Advances in Modeling Earth Systems, 16, https://doi.org/10.1029/2024ms004314, 2024.

Efstathiou, G. A. and Beare, R. J.: Quantifying and improving sub-grid diffusion in the boundary-layer grey zone, Quarterly Journal of the Royal Meteorological Society, 141, 3006–3017, https://doi.org/10.1002/qj.2585, 2015.

Faloona, I., Lenschow, D. H., Campos, T., Stevens, B., van Zanten, M., Blomquist, B., Thornton, D., Bandy, A., and Gerber, H.: Observations of Entrainment in Eastern Pacific Marine Stratocumulus Using Three Conserved Scalars, Journal of the Atmospheric Sciences, 62, 3268–3285, https://doi.org/10.1175/jas3541.1, 2005.

Feingold, G., Goren, T., and Yamaguchi, T.: Quantifying albedo susceptibility biases in shallow clouds, Atmospheric Chemistry and Physics, 22, 3303–3319, https://doi.org/10.5194/acp-22-3303-2022, 2022.

Fons, E., Runge, J., Neubauer, D., and Lohmann, U.: Stratocumulus adjustments to aerosol perturbations disentangled with a causal approach, npj Climate and Atmospheric Science, 6, https://doi.org/10.1038/s41612-023-00452-w, 2023.

Forster, P., Storelvmo, T., Armour, K., Collins, W., Dufresne, J.-L., Frame, D., Lunt, D. J., Mauritsen, T., Palmer, M. D., Watanabe, M., Wild, M., and Zhang, H.: The Earth's Energy Budget, Climate Feedbacks, and Climate Sensitivity, in: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, pp. 923–1054, Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, https://doi.org/10.1017/9781009157896.009, 2021.

Gettelman, A.: Putting the clouds back in aerosol–cloud interactions, Atmospheric Chemistry and Physics, 15, 12 397–12 411, https://doi.org/10.5194/acp-15-12397-2015, 2015.

Gettelman, A. and Morrison, H.: Advanced Two-Moment Bulk Microphysics for Global Models. Part I: Off-Line Tests and Comparison with Other Schemes, Journal of Climate, 28, 1268–1287, https://doi.org/10.1175/jcli-d-14-00102.1, 2015.

Ghan, S., Wang, M., Zhang, S., Ferrachat, S., Gettelman, A., Griesfeller, J., Kipling, Z., Lohmann, U., Morrison, H., Neubauer, D., Partridge, D. G., Stier, P., Takemura, T., Wang, H., and Zhang, K.: Challenges in constraining anthropogenic aerosol effects on cloud radiative forcing using present-day spatiotemporal variability, Proceedings of the National Academy of Sciences, 113, 5804–5811, https://doi.org/10.1073/pnas.1514036113, 2016.

Glassmeier, F. and Feingold, G.: Network approach to patterns in stratocumulus clouds, Proceedings of the National Academy of Sciences, 114, 10 578–10 583, https://doi.org/10.1073/pnas.1706495114, 2017.

Glassmeier, F., Hoffmann, F., Johnson, J. S., Yamaguchi, T., Carslaw, K. S., and Feingold, G.: An emulator approach to stratocumulus susceptibility, Atmospheric Chemistry and Physics, 19, 10 191–10 203, https://doi.org/10.5194/acp-19-10191-2019, 2019.

Glassmeier, F., Hoffmann, F., Johnson, J. S., Yamaguchi, T., Carslaw, K. S., and Feingold, G.: Aerosol-cloud-climate cooling overestimated by ship-track data, Science, 371, 485–489, https://doi.org/10.1126/science.abd3980, 2021.

Goren, T., Kazil, J., Hoffmann, F., Yamaguchi, T., and Feingold, G.: Anthropogenic Air Pollution Delays Marine Stratocumulus Breakup to Open Cells, Geophysical Research Letters, 46, 14 135–14 144, https://doi.org/10.1029/2019gl085412, 2019.

Grosvenor, D. P., Sourdeval, O., Zuidema, P., Ackerman, A., Alexandrov, M. D., Bennartz, R., Boers, R., Cairns, B., Chiu, J. C., Christensen, M., Deneke, H., Diamond, M., Feingold, G., Fridlind, A., Hünerbein, A., Knist, C., Kollias, P., Marshak, A., McCoy, D., Merk, D., Painemal, D., Rausch, J., Rosenfeld, D., Russchenberg, H., Seifert, P., Sinclair, K., Stier, P., van Diedenhoven, B., Wendisch, M., Werner, F., Wood, R., Zhang, Z., and Quaas, J.: Remote Sensing of Droplet Number Concentration in Warm Clouds: A Review of the Current State of Knowledge and Perspectives, Reviews of Geophysics, 56, 409–453, https://doi.org/10.1029/2017rg000593, 2018.

Gryspeerdt, E., Quaas, J., and Bellouin, N.: Constraining the aerosol influence on cloud fraction, Journal of Geophysical Research: Atmospheres, 121, 3566–3583, https://doi.org/10.1002/2015jd023744, 2016.

Gryspeerdt, E., Goren, T., Sourdeval, O., Quaas, J., Mülmenstädt, J., Dipu, S., Unglaub, C., Gettelman, A., and Christensen, M.: Constraining the aerosol influence on cloud liquid water path, Atmospheric Chemistry and Physics, 19, 5331–5347, https://doi.org/10.5194/acp-19-5331-2019, 2019.

Gryspeerdt, E., Goren, T., and Smith, T. W. P.: Observing the timescales of aerosol–cloud interactions in snapshot satellite images, Atmospheric Chemistry and Physics, 21, 6093–6109, https://doi.org/10.5194/acp-21-6093-2021, 2021.

Gryspeerdt, E., Glassmeier, F., Feingold, G., Hoffmann, F., and Murray-Watson, R. J.: Observing short-timescale cloud development to constrain aerosol–cloud interactions, Atmospheric Chemistry and Physics, 22, 11 727–11 738, https://doi.org/10.5194/acp-22-11727-2022, 2022a.

Gryspeerdt, E., McCoy, D. T., Crosbie, E., Moore, R. H., Nott, G. J., Painemal, D., Small-Griswold, J., Sorooshian, A., and Ziemba, L.: The impact of sampling strategy on the cloud droplet number concentration estimated from satellite data, Atmospheric Measurement Techniques, 15, 3875–3892, https://doi.org/10.5194/amt-15-3875-2022, 2022b.

Hannah, W. M., Bradley, A. M., Guba, O., Tang, Q., Golaz, J.-C., and Wolfe, J.: Separating Physics and Dynamics Grids for Improved Computational Efficiency in Spectral Element Earth System Models, Journal of Advances in Modeling Earth Systems, 13, https://doi.org/10.1029/2020ms002419, 2021.

Hill, A. A., Feingold, G., and Jiang, H.: The Influence of Entrainment and Mixing Assumption on Aerosol–Cloud Interactions in Marine Stratocumulus, Journal of the Atmospheric Sciences, 66, 1450–1464, https://doi.org/10.1175/2008jas2909.1, 2009.

Hoffmann, F. and Feingold, G.: Entrainment and Mixing in Stratocumulus: Effects of a New Explicit Subgrid-Scale Scheme for Large-Eddy Simulations with Particle-Based Microphysics, Journal of the Atmospheric Sciences, 76, 1955–1973, https://doi.org/10.1175/jas-d-18-0318.1, 2019.

Hoffmann, F., Glassmeier, F., Yamaguchi, T., and Feingold, G.: Liquid Water Path Steady States in Stratocumulus: Insights from Process-Level Emulation and Mixed-Layer Theory, Journal of the Atmospheric Sciences, 77, 2203–2215, https://doi.org/10.1175/jas-d-19-0241.1, 2020.

Hoffmann, F., Glassmeier, F., and Feingold, G.: The impact of aerosol on cloud water: a heuristic perspective, Atmospheric Chemistry and Physics, 24, 13 403–13 412, https://doi.org/10.5194/acp-24-13403-2024, 2024.

Hoffmann, F., Chen, Y.-S., and Feingold, G.: On the processes determining the slope of cloud water adjustments in weakly and non-precipitating stratocumulus, Atmospheric Chemistry and Physics, 25, 8657–8670, https://doi.org/10.5194/acp-25-8657-2025, 2025.

Honnert, R., Masson, V., and Couvreux, F.: A Diagnostic for Evaluating the Representation of Turbulence in Atmospheric Models at the Kilometric Scale, Journal of the Atmospheric Sciences, 68, 3112–3131, https://doi.org/10.1175/jas-d-11-061.1, 2011.

Honnert, R., Efstathiou, G. A., Beare, R. J., Ito, J., Lock, A., Neggers, R., Plant, R. S., Shin, H. H., Tomassini, L., and Zhou, B.: The Atmospheric Boundary Layer and the "Gray Zone" of Turbulence: A Critical Review, Journal of Geophysical Research: Atmospheres, 125, https://doi.org/10.1029/2019jd030317, 2020.

Jia, H. and Quaas, J.: Nonlinearity of the cloud response postpones climate penalty of mitigating air pollution in polluted regions, Nature Climate Change, 13, 943–950, https://doi.org/10.1038/s41558-023-01775-5, 2023.

Jia, H., Ma, X., Quaas, J., Yin, Y., and Qiu, T.: Is positive correlation between cloud droplet effective radius and aerosol optical depth over land due to retrieval artifacts or real physical processes?, Atmospheric Chemistry and Physics, 19, 8879–8896, https://doi.org/10.5194/acp-19-8879-2019, 2019.

Jia, H., Ma, X., Yu, F., and Quaas, J.: Significant underestimation of radiative forcing by aerosol–cloud interactions derived from satellite-based methods, Nature Communications, 12, https://doi.org/10.1038/s41467-021-23888-1, 2021.

Jia, H., Quaas, J., Gryspeerdt, E., Böhm, C., and Sourdeval, O.: Addressing the difficulties in quantifying droplet number response to aerosol from satellite observations, Atmospheric Chemistry and Physics, 22, 7353–7372, https://doi.org/10.5194/acp-22-7353-2022, 2022.

Jones, C. R., Bretherton, C. S., and Leon, D.: Coupled vs. decoupled boundary layers in VOCALS-REx, Atmospheric Chemistry and Physics, 11, 7143–7153, https://doi.org/10.5194/acp-11-7143-2011, 2011.

Jones, C. R., Bretherton, C. S., and Blossey, P. N.: Fast stratocumulus time scale in mixed layer model and large eddy simulation, Journal of Advances in Modeling Earth Systems, 6, 206–222, https://doi.org/10.1002/2013ms000289, 2014.

Kazil, J., Wang, H., Feingold, G., Clarke, A. D., Snider, J. R., and Bandy, A. R.: Modeling chemical and aerosol processes in the transition from closed to open cells during VOCALS-REx, Atmospheric Chemistry and Physics, 11, 7491–7514, https://doi.org/10.5194/acp-11-7491-2011, 2011.

Kazil, J., Yamaguchi, T., and Feingold, G.: Mesoscale organization, entrainment, and the properties of a closed-cell stratocumulus cloud, Journal of Advances in Modeling Earth Systems, 9, 2214–2229, https://doi.org/10.1002/2017ms001072, 2017.

Khairoutdinov, M. and Kogan, Y.: A New Cloud Physics Parameterization in a Large-Eddy Simulation Model of Marine Stratocumulus, Monthly Weather Review, 128, 229–243, https://doi.org/10.1175/1520-0493(2000)128<0229:ancppi>2.0.co;2, 2000.

Kokkola, H., Tonttila, J., Calderón, S. M., Romakkaniemi, S., Lipponen, A., Peräkorpi, A., Mielonen, T., Gryspeerdt, E., Virtanen, T. H., Kolmonen, P., and Arola, A.: Model analysis of biases in the satellite-diagnosed aerosol effect on the cloud liquid water path, Atmospheric Chemistry and Physics, 25, 1533–1543, https://doi.org/10.5194/acp-25-1533-2025, 2025.

Lilly, D. K.: Models of cloud-topped mixed layers under a strong inversion, Quarterly Journal of the Royal Meteorological Society, 94, 292–309, https://doi.org/10.1002/qj.49709440106, 1968.

Mahfouz, N., Mülmenstädt, J., and Burrows, S.: Present-day correlations are insufficient to predict cloud albedo change by anthropogenic aerosols in E3SM v2, Atmospheric Chemistry and Physics, 24, 7253–7260, https://doi.org/10.5194/acp-24-7253-2024, 2024.

Mahfouz, N., Beydoun, H., Mülmenstädt, J., Keen, N., Varble, A. C., Bertagna, L., Bogenschutz, P., Bradley, A., Christensen, M. W., Clevenger, T. C., Donahue, A., Fast, J., Foucar, J., Golaz, J.-C., Guba, O., Hannah, W., Hillman, B., Jacob, R., Lin, W., Ma, P.-L., Qian, Y., Singh, B., Terai, C., Wang, H., Wu, M., Zhang, K., Gettelman, A., Taylor, M., Leung, L. R., Caldwell, P., and Burrows, S.: Prescribing the aerosol effective radiative forcing in the Simple Cloud-Resolving E3SM Atmosphere Model v1, Atmospheric Chemistry and Physics, 25, 15 105–15 120, https://doi.org/10.5194/acp-25-15105-2025, 2025.

Matheou, G. and Teixeira, J.: Sensitivity to Physical and Numerical Aspects of Large-Eddy Simulation of Stratocumulus, Monthly Weather Review, 147, 2621–2639, https://doi.org/10.1175/mwr-d-18-0294.1, 2019.

McComiskey, A. and Feingold, G.: The scale problem in quantifying aerosol indirect effects, Atmospheric Chemistry and Physics, 12, 1031–1049, https://doi.org/10.5194/acp-12-1031-2012, 2012.

Mellado, J. P., Bretherton, C. S., Stevens, B., and Wyant, M. C.: DNS and LES for Simulating Stratocumulus: Better Together, Journal of Advances in Modeling Earth Systems, 10, 1421–1438, https://doi.org/10.1029/2018ms001312, 2018.

Michibata, T., Suzuki, K., Sato, Y., and Takemura, T.: The source of discrepancies in aerosol–cloud–precipitation interactions between GCM and A-Train retrievals, Atmospheric Chemistry and Physics, 16, 15 413–15 424, https://doi.org/10.5194/acp-16-15413-2016, 2016.

Morrison, H. and Gettelman, A.: A New Two-Moment Bulk Stratiform Cloud Microphysics Scheme in the Community Atmosphere Model, Version 3 (CAM3). Part I: Description and Numerical Tests, Journal of Climate, 21, 3642–3659, https://doi.org/10.1175/2008jcli2105.1, 2008.

Morrison, H. and Milbrandt, J. A.: Parameterization of Cloud Microphysics Based on the Prediction of Bulk Ice Particle Properties. Part I: Scheme Description and Idealized Tests, Journal of the Atmospheric Sciences, 72, 287–311, https://doi.org/10.1175/jas-d-14-0065.1, 2015.

Morrison, H., van Lier-Walqui, M., Fridlind, A. M., Grabowski, W. W., Harrington, J. Y., Hoose, C., Korolev, A., Kumjian, M. R., Milbrandt, J. A., Pawlowska, H., Posselt, D. J., Prat, O. P., Reimel, K. J., Shima, S.-I., van Diedenhoven, B., and Xue, L.: Confronting the Challenge of Modeling Cloud and Precipitation Microphysics, Journal of Advances in Modeling Earth Systems, 12, https://doi.org/10.1029/2019ms001689, 2020.

Mülmenstädt, J. and Feingold, G.: The Radiative Forcing of Aerosol–Cloud Interactions in Liquid Clouds: Wrestling and Embracing Uncertainty, Current Climate Change Reports, 4, 23–40, https://doi.org/10.1007/s40641-018-0089-y, 2018.

Mülmenstädt, J., Gryspeerdt, E., Salzmann, M., Ma, P.-L., Dipu, S., and Quaas, J.: Separating radiative forcing by aerosol–cloud interactions and rapid cloud adjustments in the ECHAM–HAMMOZ aerosol–climate model using the method of partial radiative perturbations, Atmospheric Chemistry and Physics, 19, 15 415–15 429, https://doi.org/10.5194/acp-19-15415-2019, 2019.

Mülmenstädt, J., Nam, C., Salzmann, M., Kretzschmar, J., L'Ecuyer, T. S., Lohmann, U., Ma, P.-L., Myhre, G., Neubauer, D., Stier, P., Suzuki, K., Wang, M., and Quaas, J.: Reducing the aerosol forcing uncertainty using observational constraints on warm rain processes, Science Advances, 6, https://doi.org/10.1126/sciadv.aaz6433, 2020.

Mülmenstädt, J., Ackerman, A. S., Fridlind, A. M., Huang, M., Ma, P.-L., Mahfouz, N., Bauer, S. E., Burrows, S. M., Christensen, M. W., Dipu, S., Gettelman, A., Leung, L. R., Tornow, F., Quaas, J., Varble, A. C., Wang, H., Zhang, K., and Zheng, Y.: Can general circulation models (GCMs) represent cloud liquid water path adjustments to aerosol–cloud interactions?, Atmospheric Chemistry and Physics, 24, 13 633–13 652, https://doi.org/10.5194/acp-24-13633-2024, 2024a.

Mülmenstädt, J., Gryspeerdt, E., Dipu, S., Quaas, J., Ackerman, A. S., Fridlind, A. M., Tornow, F., Bauer, S. E., Gettelman, A., Ming, Y., Zheng, Y., Ma, P.-L., Wang, H., Zhang, K., Christensen, M. W., Varble, A. C., Leung, L. R., Liu, X., Neubauer, D., Partridge, D. G., Stier, P., and Takemura, T.: General circulation models simulate negative liquid water path–droplet number correlations, but anthropogenic aerosols still increase simulated liquid water path, Atmospheric Chemistry and Physics, 24, 7331–7345, https://doi.org/10.5194/acp-24-7331-2024, 2024b.

Pincus, R., Mlawer, E. J., and Delamere, J. S.: Balancing Accuracy, Efficiency, and Flexibility in Radiation Calculations for Dynamical Models, Journal of Advances in Modeling Earth Systems, 11, 3074–3089, https://doi.org/10.1029/2019ms001621, 2019.

Platnick, S. and Twomey, S.: Determining the Susceptibility of Cloud Albedo to Changes in Droplet Concentration with the Advanced Very High Resolution Radiometer, Journal of Applied Meteorology, 33, 334–347, https://doi.org/10.1175/1520-0450(1994)033<0334:dtsoca>2.0.co;2, 1994.

Possner, A., Eastman, R., Bender, F., and Glassmeier, F.: Deconvolution of boundary layer depth and aerosol constraints on cloud water path in subtropical stratocumulus decks, Atmospheric Chemistry and Physics, 20, 3609–3621, https://doi.org/10.5194/acp-20-3609-2020, 2020.

Pressel, K. G., Mishra, S., Schneider, T., Kaul, C. M., and Tan, Z.: Numerics and subgrid-scale modeling in large eddy simulations of stratocumulus clouds, Journal of Advances in Modeling Earth Systems, 9, 1342–1365, https://doi.org/10.1002/2016ms000778, 2017.

Quaas, J., Ming, Y., Menon, S., Takemura, T., Wang, M., Penner, J. E., Gettelman, A., Lohmann, U., Bellouin, N., Boucher, O., Sayer, A. M., Thomas, G. E., McComiskey, A., Feingold, G., Hoose, C., Kristjánsson, J. E., Liu, X., Balkanski, Y., Donner, L. J., Ginoux, P. A., Stier, P., Grandey, B., Feichter, J., Sednev, I., Bauer, S. E., Koch, D., Grainger, R. G., Kirkevåg, A., Iversen, T., Seland, Ø., Easter, R., Ghan, S. J., Rasch, P. J., Morrison, H., Lamarque, J.-F., Iacono, M. J., Kinne, S., and Schulz, M.: Aerosol indirect effects – general circulation model intercomparison and evaluation with satellite data, Atmospheric Chemistry and Physics, 9, 8697–8717, https://doi.org/10.5194/acp-9-8697-2009, 2009.

Quaas, J., Arola, A., Cairns, B., Christensen, M., Deneke, H., Ekman, A. M. L., Feingold, G., Fridlind, A., Gryspeerdt, E., Hasekamp, O., Li, Z., Lipponen, A., Ma, P.-L., Mülmenstädt, J., Nenes, A., Penner, J. E., Rosenfeld, D., Schrödner, R., Sinclair, K., Sourdeval, O., Stier, P., Tesche, M., van Diedenhoven, B., and Wendisch, M.: Constraining the Twomey effect from satellite observations: issues and perspectives, Atmospheric Chemistry and Physics, 20, 15 079–15 099, https://doi.org/10.5194/acp-20-15079-2020, 2020.

Sandu, I., Brenguier, J.-L., Geoffroy, O., Thouron, O., and Masson, V.: Aerosol Impacts on the Diurnal Cycle of Marine Stratocumulus, Journal of the Atmospheric Sciences, 65, 2705–2718, https://doi.org/10.1175/2008jas2451.1, 2008.

Sato, Y., Goto, D., Michibata, T., Suzuki, K., Takemura, T., Tomita, H., and Nakajima, T.: Aerosol effects on cloud water amounts were successfully simulated by a global cloud-system resolving model, Nature Communications, 9, https://doi.org/10.1038/s41467-018-03379-6, 2018.

Savic-Jovcic, V. and Stevens, B.: The Structure and Mesoscale Organization of Precipitating Stratocumulus, Journal of the Atmospheric Sciences, 65, 1587–1605, https://doi.org/10.1175/2007jas2456.1, 2008.

Schneider, T., Teixeira, J., Bretherton, C. S., Brient, F., Pressel, K. G., Schär, C., and Siebesma, A. P.: Climate goals and computing the future of clouds, Nature Climate Change, 7, 3–5, https://doi.org/10.1038/nclimate3190, 2017.

Seifert, A., Heus, T., Pincus, R., and Stevens, B.: Large-eddy simulation of the transient and near-equilibrium behavior of precipitating shallow convection, Journal of Advances in Modeling Earth Systems, 7, 1918–1937, https://doi.org/10.1002/2015ms000489, 2015.

Small, J. D., Chuang, P. Y., Feingold, G., and Jiang, H.: Can aerosol decrease cloud lifetime?, Geophysical Research Letters, 36, https://doi.org/10.1029/2009gl038888, 2009.

Stanford, M., Mahfouz, N., Mülmenstädt, J., Mikkelsen, A., and Burrows, S.: Towards Resolving Equifinality in Aerosol-Cloud Radiative Forcing Through Process-Level Constraints, EGUsphere [preprint], https://doi.org/10.5194/egusphere-2026-4488, 2026.

Stevens, B.: Entrainment in stratocumulus-topped mixed layers, Quarterly Journal of the Royal Meteorological Society, 128, 2663–2690, https://doi.org/10.1256/qj.01.202, 2002.

Stevens, B. and Feingold, G.: Untangling aerosol effects on clouds and precipitation in a buffered system, Nature, 461, 607–613, https://doi.org/10.1038/nature08281, 2009.

Stevens, B., Cotton, W. R., Feingold, G., and Moeng, C.-H.: Large-Eddy Simulations of Strongly Precipitating, Shallow, Stratocumulus-Topped Boundary Layers, Journal of the Atmospheric Sciences, 55, 3616–3638, https://doi.org/10.1175/1520-0469(1998)055<3616:lesosp>2.0.co;2, 1998.

Stevens, B., Moeng, C.-H., and Sullivan, P. P.: Large-Eddy Simulations of Radiatively Driven Convection: Sensitivities to the Representation of Small Scales, Journal of the Atmospheric Sciences, 56, 3963–3984, https://doi.org/10.1175/1520-0469(1999)056<3963:lesord>2.0.co;2, 1999.

Stevens, B., Lenschow, D. H., Vali, G., Gerber, H., Bandy, A., Blomquist, B., Brenguier, J.-L., Bretherton, C. S., Burnet, F., Campos, T., Chai, S., Faloona, I., Friesen, D., Haimov, S., Laursen, K., Lilly, D. K., Loehrer, S. M., Malinowski, S. P., Morley, B., Petters, M. D., Rogers, D. C., Russell, L., Savic-Jovcic, V., Snider, J. R., Straub, D., Szumowski, M. J., Takagi, H., Thornton, D. C., Tschudi, M., Twohy, C., Wetzel, M., and van Zanten, M. C.: Dynamics and chemistry of marine stratocumulus—DYCOMS-II, Bulletin of the American Meteorological Society, 84, 579–593, https://doi.org/10.1175/BAMS-84-5-579, 2003.

Stevens, B., Moeng, C.-H., Ackerman, A. S., Bretherton, C. S., Chlond, A., de Roode, S., Edwards, J., Golaz, J.-C., Jiang, H., Khairoutdinov, M., Kirkpatrick, M. P., Lewellen, D. C., Lock, A., Müller, F., Stevens, D. E., Whelan, E., and Zhu, P.: Evaluation of Large-Eddy Simulations via Observations of Nocturnal Marine Stratocumulus, Monthly Weather Review, 133, 1443–1462, https://doi.org/10.1175/mwr2930.1, 2005a.

Stevens, B., Vali, G., Comstock, K., Wood, R., van Zanten, M. C., Austin, P. H., Bretherton, C. S., and Lenschow, D. H.: POCKETS OF OPEN CELLS AND DRIZZLE IN MARINE STRATOCUMULUS, Bulletin of the American Meteorological Society, 86, 51–58, https://doi.org/10.1175/bams-86-1-51, 2005b.

Stevens, B., Satoh, M., Auger, L., Biercamp, J., Bretherton, C. S., Chen, X., Düben, P., Judt, F., Khairoutdinov, M., Klocke, D., Kodama, C., Kornblueh, L., Lin, S.-J., Neumann, P., Putman, W. M., Röber, N., Shibuya, R., Vanniere, B., Vidale, P. L., Wedi, N., and Zhou, L.: DYAMOND: the DYnamics of the Atmospheric general circulation Modeled On Non-hydrostatic Domains, Progress in Earth and Planetary Science, 6, https://doi.org/10.1186/s40645-019-0304-z, 2019.

STEVENS, B., ACQUISTAPACE, C., HANSEN, A., HEINZE, R., KLINGER, C., KLOCKE, D., RYBKA, H., SCHUBOTZ, W., WINDMILLER, J., ADAMIDIS, P., ARKA, I., BARLAKAS, V., BIERCAMP, J., BRUECK, M., BRUNE, S., BUEHLER, S. A., BURKHARDT, U., CIONI, G., COSTA-SURÓS, M., CREWELL, S., CRÜGER, T., DENEKE, H., FRIEDERICHS, P., HENKEN, C. C., Hohenegger, C., JACOB, M., JAKUB, F., KALTHOFF, N., KÖHLER, M., van LAAR, T. W., LI, P., LÖHNERT, U., MACKE, A., MADENACH, N., MAYER, B., NAM, C., NAUMANN, A. K., PETERS, K., POLL, S., QUAAS, J., RÖBER, N., ROCHETIN, N., SCHECK, L., SCHEMANN, V., SCHNITT, S., SEIFERT, A., SENF, F., SHAPKALIJEVSKI, M., SIMMER, C., SINGH, S., SOURDEVAL, O., SPICKERMANN, D., STRANDGREN, J., TESSIOT, O., VERCAUTEREN, N., VIAL, J., VOIGT, A., and ZÄNGL, G.: The Added Value of Large-eddy and Storm-resolving Models for Simulating Clouds and Precipitation, Journal of the Meteorological Society of Japan. Ser. II, 98, 395–435, https://doi.org/10.2151/jmsj.2020-021, 2020.

Stevens, D. E. and Bretherton, C. S.: Effects of resolution on the simulation of stratocumulus entrainment, Quarterly Journal of the Royal Meteorological Society, 125, 425–439, https://doi.org/10.1002/qj.49712555403, 1999.

Taylor, M. A., Guba, O., Steyer, A., Ullrich, P. A., Hall, D. M., and Eldred, C.: An Energy Consistent Discretization of the Nonhydrostatic Equations in Primitive Variables, Journal of Advances in Modeling Earth Systems, 12, https://doi.org/10.1029/2019ms001783, 2020.

Terai, C. R., Wood, R., Leon, D. C., and Zuidema, P.: Does precipitation susceptibility vary with increasing cloud thickness in marine stratocumulus?, Atmospheric Chemistry and Physics, 12, 4567–4583, https://doi.org/10.5194/acp-12-4567-2012, 2012.

Terai, C. R., Pritchard, M. S., Blossey, P., and Bretherton, C. S.: The Impact of Resolving Subkilometer Processes on Aerosol-Cloud Interactions of Low-Level Clouds in Global Model Simulations, Journal of Advances in Modeling Earth Systems, 12, https://doi.org/10.1029/2020ms002274, 2020.

Toll, V., Christensen, M., Quaas, J., and Bellouin, N.: Weak average liquid-cloud-water response to anthropogenic aerosols, Nature, 572, 51–55, https://doi.org/10.1038/s41586-019-1423-9, 2019.

Turton, J. D. and Nicholls, S.: A Study of the Diurnal Variation of Stratocumulus Using A Multiple Mixed Layer Model, Quarterly Journal of the Royal Meteorological Society, 113, 969–1009, https://doi.org/10.1002/qj.49711347712, 1987.

Twomey, S.: Pollution and the planetary albedo, Atmospheric Environment (1967), 8, 1251–1256, https://doi.org/10.1016/0004-6981(74)90004-3, 1974.

Twomey, S.: The Influence of Pollution on the Shortwave Albedo of Clouds, Journal of the Atmospheric Sciences, 34, 1149–1152, https://doi.org/10.1175/1520-0469(1977)034<1149:tiopot>2.0.co;2, 1977.

Uchida, J., Bretherton, C. S., and Blossey, P. N.: The sensitivity of stratocumulus-capped mixed layers to cloud droplet concentration: do LES and mixed-layer models agree?, Atmospheric Chemistry and Physics, 10, 4097–4109, https://doi.org/10.5194/acp-10-4097-2010, 2010.

van der Dussen, J. J., de Roode, S. R., and Siebesma, A. P.: Factors Controlling Rapid Stratocumulus Cloud Thinning, Journal of the Atmospheric Sciences, 71, 655–664, https://doi.org/10.1175/jas-d-13-0114.1, 2014.

vanZanten, M. C. and Stevens, B.: Observations of the Structure of Heavily Precipitating Marine Stratocumulus, Journal of the Atmospheric Sciences, 62, 4327–4342, https://doi.org/10.1175/jas3611.1, 2005.

vanZanten, M. C., Stevens, B., Vali, G., and Lenschow, D. H.: Observations of Drizzle in Nocturnal Marine Stratocumulus, Journal of the Atmospheric Sciences, 62, 88–106, https://doi.org/10.1175/jas-3355.1, 2005.

Wang, M., Ghan, S., Ovchinnikov, M., Liu, X., Easter, R., Kassianov, E., Qian, Y., and Morrison, H.: Aerosol indirect effects in a multi-scale aerosol-climate model PNNL-MMF, Atmospheric Chemistry and Physics, 11, 5431–5455, https://doi.org/10.5194/acp-11-5431-2011, 2011.

Wang, S., Wang, Q., and Feingold, G.: Turbulence, Condensation, and Liquid Water Transport in Numerically Simulated Nonprecipitating Stratocumulus Clouds, Journal of the Atmospheric Sciences, 60, 262–278, https://doi.org/10.1175/1520-0469(2003)060<0262:tcalwt>2.0.co;2, 2003.

Wood, R.: Drizzle in Stratiform Boundary Layer Clouds. Part II: Microphysical Aspects, Journal of the Atmospheric Sciences, 62, 3034–3050, https://doi.org/10.1175/jas3530.1, 2005.

Wood, R.: Cancellation of Aerosol Indirect Effects in Marine Stratocumulus through Cloud Thinning, Journal of the Atmospheric Sciences, 64, 2657–2669, https://doi.org/10.1175/jas3942.1, 2007.

Wood, R.: Stratocumulus Clouds, Monthly Weather Review, 140, 2373–2423, https://doi.org/10.1175/mwr-d-11-00121.1, 2012.

Wood, R. and Hartmann, D. L.: Spatial Variability of Liquid Water Path in Marine Low Cloud: The Importance of Mesoscale Cellular Convection, Journal of Climate, 19, 1748–1764, https://doi.org/10.1175/jcli3702.1, 2006.

Wyant, M. C., Bretherton, C. S., Chlond, A., Griffin, B. M., Kitagawa, H., Lappen, C.-L., Larson, V. E., Lock, A., Park, S., de Roode, S. R., Uchida, J., Zhao, M., and Ackerman, A. S.: A single-column model intercomparison of a heavily drizzling stratocumulus-topped boundary layer, Journal of Geophysical Research: Atmospheres, 112, https://doi.org/10.1029/2007jd008536, 2007.

Wyngaard, J. C.: Toward Numerical Modeling in the "Terra Incognita", Journal of the Atmospheric Sciences, 61, 1816–1826, https://doi.org/10.1175/1520-0469(2004)061<1816:tnmitt>2.0.co;2, 2004.

Yamaguchi, T., Feingold, G., and Kazil, J.: Stratocumulus to Cumulus Transition by Drizzle, Journal of Advances in Modeling Earth Systems, 9, 2333–2349, https://doi.org/10.1002/2017ms001104, 2017.

Zhang, J. and Feingold, G.: Distinct regional meteorological influences on low-cloud albedo susceptibility over global marine stratocumulus regions, Atmospheric Chemistry and Physics, 23, 1073–1090, https://doi.org/10.5194/acp-23-1073-2023, 2023.

Zhang, J. and Zuidema, P.: The diurnal cycle of the smoky marine boundary layer observed during August in the remote southeast Atlantic, Atmospheric Chemistry and Physics, 19, 14 493–14 516, https://doi.org/10.5194/acp-19-14493-2019, 2019.

Zhang, J., Chen, Y.-S., Yamaguchi, T., and Feingold, G.: Cloud water adjustments to aerosol perturbations are buffered by solar heating in non-precipitating marine stratocumuli, Atmospheric Chemistry and Physics, 24, 10 425–10 440, https://doi.org/10.5194/acp-24-10425-2024, 2024.

Zhou, X. and Bretherton, C. S.: Simulation of Mesoscale Cellular Convection in Marine Stratocumulus: 2. Nondrizzling Conditions, Journal of Advances in Modeling Earth Systems, 11, 3–18, https://doi.org/10.1029/2018ms001448, 2019.

Zhou, X., Ackerman, A. S., Fridlind, A. M., and Kollias, P.: Simulation of Mesoscale Cellular Convection in Marine Stratocumulus. Part I: Drizzling Conditions, Journal of the Atmospheric Sciences, 75, 257–274, https://doi.org/10.1175/jas-d-17-0070.1, 2018.

Zhu, P., Bretherton, C. S., Köhler, M., Cheng, A., Chlond, A., Geng, Q., Austin, P., Golaz, J.-C., Lenderink, G., Lock, A., and Stevens, B.: Intercomparison and Interpretation of Single-Column Model Simulations of a Nocturnal Stratocumulus-Topped Marine Boundary Layer, Monthly Weather Review, 133, 2741–2758, https://doi.org/10.1175/mwr2997.1, 2005.